

Enhancing X-Ray Images from XMM-Newton

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Abstract

This thesis explores the application of Artificial Intelligence (AI) to enhance X-ray images captured by the X-ray Multi Mirror (XMM)-Newton satellite, focusing on Super-Resolution (SR) and Denoising (DN) techniques. Building on the work of Sweere et al. [1], we employ the same models they used and additionally introduce DRCT [2] for SR and Restormer [3] for DN. Our models, trained on a custom dataset, outperform Sweere et al.'s [1] models in the task of SR, both quantitatively and qualitatively. The only exception is DRCT, which, due to a short training period, does not perform as well in visual evaluations.

In the task of DN, our models do not achieve the same quantitative performance as Sweere et al.'s [1]. However, we produce superior visual results on real XMM-Newton observations, highlighting the practical applicability of our approach. On the simulated dataset, our visual performance is mixed. While Restormer [3] does not deliver satisfactory results, likely due to the limited training period, the overall improvements demonstrate the potential of modern AI techniques in enhancing astronomical X-ray imaging.

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1 Introduction

In the field of astronomy, the ability to capture and analyse high-quality images is essential for understanding the universe. X-ray astronomy, in particular, provides a unique window into some of the most energetic processes and objects in space, such as black holes, neutron stars, and supernovae. The XMM-Newton satellite, launched by the European Space Agency, has been instrumental in collecting X-ray data from these phenomena, contributing significantly to our understanding of the cosmos. However, like all observational instruments, the XMM-Newton satellite faces limitations in image resolution and noise, which can hinder the accuracy and depth of scientific analysis.

Recent advances in Artificial Intelligence (AI) have opened up new possibilities for improving the quality of images from a variety of sources, including astronomical observations. By applying AI-driven techniques such as Super-Resolution (SR) and Denoising (DN), it is possible to enhance the clarity and detail of images, offering astronomers more accurate data for their research. Despite the potential of AI in this area, there has been limited exploration of these techniques within the context of X-ray imaging. Most existing work on SR and DN focuses on common optical images, while the few studies that do address astronomical X-ray data rely on older models or aim for different objectives.

This thesis seeks to bridge that gap by exploring the application of modern AI models to enhance X-ray images captured by the XMM-Newton satellite. Our approach involves creating a custom dataset and applying cutting-edge AI techniques for image restoration. The thesis is structured as follows: First, we provide the necessary background on key topics in chapter 2. Then, we review related work in chapter 3, focusing on SR and DN models for both common and X-ray images. The subsequent chapters detail the dataset we developed, the models we employed, and their performance evaluation. Finally, the thesis concludes with a summary of findings and suggestions for future work in this emerging field.

2 Background

This chapter lays the groundwork for understanding the core concepts and technologies applied throughout this thesis. We begin in section 2.1 with an overview of methods used to simulate the Universe, focusing on its components, cosmological models, initial conditions, and the significance of dark matter and hydrodynamical simulations. In section 2.2, we introduce the Illustris Project, a pivotal cosmological simulation effort, concentrating on the second data release, which forms the basis of this work.

Next, in section 2.3, we focus on X-ray imaging, examining the detectors used in X-ray astronomy, especially Charge-Coupled Devices (CCDs), which are essential for capturing astronomical data. Section 2.4 follows with a discussion of the XMM-Newton satellite and its instruments, a pivotal tool in modern X-ray astrophysics.

In section 2.5, we explore the Simulation of X-ray Telescopes (SIXTE) software, which simulates the behaviour of X-ray telescopes, connecting theoretical models with observational data. The chapter then transitions to Artificial Intelligence (AI) in section 2.6, offering a historical overview of AI's development, from early perceptrons to modern advances such as Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), transformers, and autoencoders. Finally, section 2.7 covers image restoration techniques, including Image Quality Assessment (IQA), Super-Resolution (SR), and Denoising (DN).

2.1 Simulating the Universe

Salmon [4] and Baumann [5] describe how the density fluctuations in the early Universe eventually gave rise to the cosmic structures we see today. While the large-scale gravitational clustering of matter can be modelled analytically, smaller scales are far more complex and require numerical simulations to capture their highly non-linear nature. Baumann [5] highlights that dark matter formed a vast web-like structure with dense nodes linked by filaments. In these regions of high dark matter density, baryonic gas accumulated, leading to star formation and, around a billion years after the Big Bang, the emergence of galaxies. Despite significant advancements, Genel et al. [6] and Vogelsberger et al. [7] emphasise that the precise processes behind galaxy formation remain largely unknown due to the complex non-linear dynamics combined with the numerous scales and physical forces shaping galaxies. As a result, cosmological simulations have become the primary tool for predicting galaxies' abundance, distribution, internal properties, and evolution. These simulations track the main components of the Universe, (cold) dark matter, dark energy, and baryonic matter, starting from cosmological initial conditions.

2.1.1 Components of the Universe

The term *dark matter* refers to components whose nature remains largely unknown. It is invisible, and its origins are yet to be fully understood. Zwicky [8], according to Baumann [5], first proposed the existence of dark matter in 1933 after observing that galaxies within the Coma cluster were moving faster than expected based on the gravitational influence of visible matter alone. To account for this discrepancy, Zwicky [8], as cited by Baumann [5], suggested the presence of additional, unseen matter necessary to hold the cluster together. Baumann [5] notes that the most compelling evidence for dark matter arises from observations of the Cosmic Microwave Background (CMB), though identifying a microscopic explanation remains a subject of ongoing research.

On large scales, all dark matter models are characterised as pressureless fluids, hence the term *cold dark matter*. This form of matter is central to the standard model of cosmology. However, in the 1980s, the concept of a Universe composed solely of matter encountered a crisis, as highlighted by Baumann [5]. Two significant discrepancies emerged: firstly, the estimated age of a matter-only universe appeared shorter than the ages of the oldest stars, an impossible contradiction. Secondly, the total matter density was insufficient for a spatially flat universe, which conflicted with theoretical predictions from inflationary cosmology. To resolve this „missing energy“ problem, a new form of energy - *dark energy* - was proposed. Although observational evidence for dark energy is robust, its exact nature remains a mystery. This discovery triggered a new challenge in cosmology, as dark energy is now believed to make up 70 % of the Universe’s total energy. One possible candidate for this enigmatic force is the energy density of empty space itself.

Baumann [5] further explains that the scientific community has a better understanding of *baryons*, the final major component of the Universe. Baryons, which consist of atoms and electrons, form all visible matter. However, despite being the material of everything we can observe, baryons account for only 5 % of the Universe’s total mass, and their origins and formation remain only partially understood.

Baumann [5] offers a detailed explanation of the CMB. Around 380 000 years after the Big Bang, the Universe had cooled enough for the first stable atoms to form. At this point, light no longer scattered off free electrons in the plasma and could travel freely through space, having continued to do so through the Universe ever since. Stretched over 13.8 billion years of cosmic expansion, this first light is detected today as faint microwave radiation, the Cosmic Microwave Background (CMB). It contains subtle variations in intensity, reflecting matter density fluctuations in the early Universe. These fluctuation patterns carry vital information about the conditions in the primordial Universe.

The stretching of this light over time, referred to as redshift, is commonly denoted by the symbol z , as explained by Baumann [5]. Redshift refers to the increase in a light wave’s observed wavelength due to the Universe’s expansion, with red light having a longer wavelength than blue. Baumann [5] further clarifies that the wavelength of light λ is proportional to the cosmological scale factor a and can be expressed as shown in eq. (2.1).

$$1 + z = \frac{1}{a(t_1)}. \quad (2.1)$$

For example, light from a galaxy at redshift $z = 1$ was emitted when the Universe was half its current size.

2.1.2 Cosmological Models and Initial Conditions

All simulations are based on a cosmological model and initial conditions, as stated by Vogelsberger et al. [7]. According to Vogelsberger et al. [9], the standard model, known as

the standard Λ Cold Dark Matter (Λ CDM) cosmological model, is the foundation for most simulations.

Carroll [10] explains that the symbol Λ represents the cosmological constant, which Einstein initially introduced to achieve a static Universe where the scale factor a does not change over time. Einstein intended to incorporate Mach's principle, which asserts that matter determines inertia, and to account for the astronomical data understood at the time. Consequently, he introduced Λ as a new free parameter in his equations. Baumann [5] adds that although the observed value of Λ is significantly smaller than all particle physics scales, making it challenging to interpret from a more fundamental perspective, it has been identified as a measure of the energy density of the vacuum, the state of lowest energy. This identification allows for considering the scales of various contributions to the cosmological constant.

Perivolaropoulos et al. [11] give six additional assumptions made by the Λ CDM model. Firstly, *General Relativity* is the theory that accurately describes gravity on cosmological scales, operating within the framework of the *Cosmological Principle*, which asserts that the Universe is homogeneous and isotropic in both space and matter at sufficiently large scales. To model the Universe under these assumptions, *six independent parameters* are typically used: the energy densities of baryons (ω_b) and cold dark matter (ω_c), the angular diameter distance to the sound horizon at the time of last scattering (θ_s), the amplitude (A_s) and tilt (n_s) of primordial scalar fluctuations, and the reionisation optical depth (τ). The *spatial part of the cosmic metric* is assumed to be flat, consistent with the Cosmological Principle, and is described by the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. Furthermore, a *primordial phase of cosmic inflation*, a period of rapid, accelerated expansion, is assumed to resolve the horizon and flatness problems. Gaussian scale-invariant primordial fluctuations were produced during this inflationary period due to quantum fluctuations in the inflationary epoch. Despite being the standard model, the Λ CDM model has flaws, comprehensively summarised by Perivolaropoulos et al. [11].

Selecting the *initial conditions* for cosmological simulations is more complex than choosing the cosmological model, as noted by Vogelsberger et al. [7]. Two commonly employed types of conditions are uniformly sampled periodic large volumes and zoom initial conditions, the latter featuring a low-resolution background realisation of the density fields surrounding a high-resolution region of interest. These initial conditions define the perturbations imposed on top of a homogeneous expanding background. Typically, the background model is assumed to be a spatially flat FLRW space-time with a specified composition of dark matter, dark energy, and baryons. Vogelsberger et al. [12] outline several parameters that characterise the initial conditions of cosmological simulations. These include the mass and energy density contributions from matter Ω_m , the cosmological constant Ω_Λ , and baryons Ω_b . Additional parameters encompass the Hubble expansion rate h at redshift $z = 0$, the spectral index n_s of the primordial power spectrum, and the root mean squared amplitude of mass fluctuations σ_8 in $8\frac{Mpc}{h}$ spheres, linearly extrapolated to redshift $z = 0$.

2.1.3 Cosmological Simulations

Vogelsberger et al. [7] note that the field of simulations has advanced considerably over the past 50 years, with increasingly sophisticated models emerging. These simulations are typically classified by their simulated volume and the type of matter they simulate. Large-scale simulations provide statistical insights into galaxies, while zoom simulations offer more detailed information on smaller structures. Simulations can also focus solely on dark matter or include both dark and ordinary matter, known as hydrodynamical simulations.

Genel et al. [6] and Vogelsberger et al. [12] state that it is commonly assumed that the evolution of dark components is predominantly governed by gravitational interactions, which

can be effectively simulated using collisionless N -body simulations. These *dark matter simulations* are relatively straightforward to model and are a standard component of all cosmological simulations. Genel et al. [6] further clarify that the behaviour of dark matter is described by the collisionless Boltzmann equation coupled with Poisson’s equation. The Boltzmann equation governs the evolution of dark matter’s phase-space density or distribution function under the influence of the collective gravitational potential defined by Poisson’s equation. Due to the high dimensionality of the Boltzmann equation, efficient numerical solution methods are challenging to implement. The commonly employed approach for addressing this issue is known as N -body methods, which derive their name from the fact that the phase-space density is sampled by an ensemble of N phase-space points. Press and Schechter [13] pioneered simulating dark matter using one such method in 1974.

In contrast to dark matter, the behaviour of baryonic matter is shaped by highly complex and sometimes poorly understood physical processes, as outlined by Genel et al. [6]. While simplifying specific physical components can reduce this complexity, it often comes at the cost of the simulations’ predictive accuracy. Including both types of matter is why researchers rely on *hydrodynamical simulations*. However, these simulations are computationally demanding due to the range of dynamics involved, including highly supersonic flows and large Reynolds numbers. Furthermore, hydrodynamical simulations face additional complexity because they must account for numerous processes that influence the formation and evolution of galaxies. Many of these processes are incorporated through so-called sub-resolution models necessary to overcome the simulations’ numerical resolution limitations. These models include gas cooling, stellar feedback, and magnetic fields.

Astrophysical gases, which the baryon component is initially comprised of, are typically modelled as ideal gases governed by the Euler equations, and according to Vogelsberger et al. [7], three main numerical methods are used to approach this: Eulerian, Lagrangian, and Arbitrary Lagrangian-Eulerian (ALE) methods. *Eulerian* methods, the traditional approach, solve the hyperbolic partial differential equations that describe ideal hydrodynamics by focusing on fixed locations in space through which fluid flows over time. Standard techniques include finite volume, finite difference, finite element, and spectral or wavelet methods. A popular mesh-free *Lagrangian* method is Smoothed Particle Hydrodynamics (SPH), which approximates fluid dynamics using sampling particles that follow the equations of motion derived from the hydrodynamical equations. This approach assumes an observer tracking an individual fluid parcel and is particularly useful for representing large density contrasts. While some cosmological simulations have employed Lagrangian mesh-based hydrodynamics schemes that rely on grid deformation techniques, these methods often suffer from mesh-tangling issues. *Arbitrary Lagrangian-Eulerian (ALE)* methods allow for flexible grid velocity. A notable example is the **AREPO** method, developed by Springel [14], which effectively avoids mesh-tangling problems and is introduced briefly in section 2.2.

Vogelsberger et al. [7] further clarify the distinction between mesh-free and mesh-based methods. In *mesh-free* approaches, no connections between nodes are required, with each node interacting directly with its neighbours. In contrast, *mesh-based* algorithms rely on predefined connections between nodes.

Genel et al. [6] identify three fundamental areas for improvement in hydrodynamical simulations. Firstly, as computational power has increased, simulations have expanded in size, allowing for finer details to be resolved and facilitating more robust numerical modelling. This enhancement also enables more thorough comparisons with observational datasets. Secondly, advancements in the scope and complexity of sub-resolution models have resulted in greater numerical robustness and a more comprehensive depiction of the processes that

shape galaxies. Lastly, introducing new numerical techniques has led to more precise solutions for the underlying equations of gravity and hydrodynamics, thereby enhancing the overall accuracy of the simulations.

2.2 The Illustris Project

The Illustris project, introduced in 2014 through a series of three papers by Genel et al. [6] and Vogelsberger et al. [9, 12], represents a cutting-edge simulation that integrates advancements in the previously mentioned research areas. According to its creators, it is the first single, self-consistent simulation capable of simultaneously predicting both large-scale statistics and small-scale galaxy properties. Furthermore, they claim that the project's findings demonstrate that the Λ CDM model accurately describes observational data across various scales and predicts a significant, scale-dependent impact of baryonic effects on dark matter distribution, with implications for future cosmological probes.

The pre-processing of the snapshots involves the identification of halos, subhalos, and their properties using extended and optimised versions of the FOF [15, 16] and SUBFIND [16, 17] algorithms, as described by Genel et al. [6] and Vogelsberger et al. [9]. Gravitational forces are calculated using a Tree-PM scheme [18], where long-range forces are determined using a particle-mesh method. In contrast, short-range forces are computed via a hierarchical `octree` algorithm, where `octree` is a tree data structure where each internal node has exactly eight children.

According to Vogelsberger et al. [12], the simulation evolves the Universe from an initial redshift of $z = 127$ to the present redshift of $z = 0$, starting approximately 12.6 million years after the Big Bang. For this, the novel moving-mesh code AREPO [14], an ALE method is employed. This method utilises a moving unstructured Voronoi tessellation in combination with a finite volume approach, and according to its creator Springel [14], the mathematical properties of the Voronoi tessellation enable the mesh to deform and change its topology continuously, avoiding the issues typically associated with traditional ALE methods, such as mesh-tangling effects. The moving-mesh approach strikes a balance between SPH and Adaptive Mesh Refinement (AMR) [19], inheriting the automatic adaptivity, geometric flexibility, and Galilean invariance of SPH while also benefiting from the high-accuracy treatment of shocks, shear waves, and fluid instabilities, as well as the low noise and absence of artificial viscosity characteristic of AMR.

Since its inception, two public data releases have been published by Nelson et al. The first set of simulations, named Illustris, was published in 2015 [20], while the second set, known as IllustrisTNG (The Next Generation (TNG)), was released in 2021 [21]. For simplicity, we will refer to the first simulations as Illustris and the subsequent ones as TNG.

2.2.1 The First Data Release

According to Vogelsberger et al. [12], the first data release has three primary and three supplementary simulations. Each simulation represents a periodic box measuring 106.5 Mpc on each side, resulting in a volume of $(106.5 \text{ Mpc})^3$. At $z = 0$, these simulations collectively contain 41 416 galaxies. Genel et al. [6] add that the primary runs also incorporate passive Monte Carlo tracer particles, enabling a Lagrangian tracking of gas flow.

Nelson et al. [20] complements the previous publications with a summary of the main simulations. The largest and most complete simulation in terms of implemented physical processes, Illustris-1, follows 1820^3 dark matter and an equivalent number of baryonic resolution elements. At $z = 0$, the smallest scale is approximately 4 pc, comparable to state-of-the-art large-scale N -body simulations which do not include baryons. Illustris-1

Ω_Λ	Ω_m	Ω_b	σ_8	n_s	H_0
0.7274	0.2726	0.0456	0.809	0.9673	$70.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$

Table 2.1: Initial conditions of Illustris as given by Vogelsberger et al. [12].

also resolves gravitational dynamics to about 710 pc at $z = 0$ while tracking the large-scale evolution in a full volume. Illustris-2 and Illustris-3 are lower resolution versions of the simulation, employing 2×910^3 and 2×455^3 resolution elements, respectively, primarily for studying resolution. The auxiliary simulations have less comprehensive physics implementation, focusing solely on simulating dark matter. Non-radiative runs exist, but their results have not been made public.

The initial conditions have been determined, according to Genel et al. [6], through a series of 100 simulations representing various random realisations of Illustris cosmology in a 106.5 Mpc volume evolved using a low-resolution Dark Matter-only simulation to $z = 0$. The selection criteria require the volume to display relative spatial homogeneity upon visual inspection and for the Dark Matter halo mass function to align closely with the theoretical mass function. The selected values are given in Vogelsberger et al. [12], according to which these are consistent with the Wilkinson Microwave Anisotropy Probe 9 measurements, and are shown in table 2.1.

As defined by van Dommelen [22], a periodic box is a simple approach to modelling large or even infinite systems by considering a finite but periodic instead of an infinite box. A particle that exits such a box through one side reenters it simultaneously through the opposite side.

Baumann [5] explains that the parallax second pc is a unit of distance derived from trigonometric principles, specifically due to the Earth’s revolution around the Sun. This movement causes slight shifts in the positions of stars, resulting in different viewing angles. A pc is defined as the distance at which the radius of Earth’s orbit subtends an angle of one arc-second, which is equal to $\frac{1}{3600}$ of one degree. Consequently, 1 pc is approximately 3.26 light years.

2.2.2 The Second Data Release

According to Nelson et al. [20], several significant discrepancies were identified in the simulated galaxy and halo populations of Illustris compared to observational constraints. Consequently, the authors cautioned against drawing definitive scientific conclusions in some cases, prompting the development of its successor: the TNG simulations.

The TNG suite consists of three simulations: TNG50, TNG100, and TNG300. The first two simulations, TNG100 and TNG300, were presented in a series of five papers by Marinacci et al. [23], Naiman et al. [24], Nelson et al. [25], Pillepich et al. [26], and Springel et al. [27]. The third and final simulation, TNG50, boasts significantly higher resolution and was introduced by Nelson et al. [28] and Pillepich et al. [29]. According to Nelson et al. [21], initial explorations of TNG, particularly in the TNG100-1 and TNG300-1 simulations, have shown no significant discrepancies when compared to observational data. Thus, there are no specific concerns regarding the use of TNG in any of the regimes where the original Illustris simulation was found to be less robust. The results of all these simulations were subsequently made public by Nelson et al. [21].

Pillepich et al. [26] and Weinberger et al. [30] highlight that the TNG simulations incorporate a comprehensive model for galaxy formation physics, enabling realistic tracking of the formation and evolution of galaxies over cosmic time. Nelson et al. [21] summarise the simulations included in TNG. The large physical volume of the largest simulation

Ω_Λ	Ω_m	Ω_b	σ_8	n_s	H_0
0.6911	0.3089	0.0456	0.8159	0.9667	$67.74 \text{ km s}^{-1} \text{ Mpc}^{-1}$

Table 2.2: Initial conditions of TNG as given by Nelson et al. [21].

Name Unit	Volume Mpc^3	N_{Gas}	N_{Tracer}	$N_{\text{DarkMatter}}$
TNG50-1	51.7^3	2160^3	1×2160^3	2160^3
TNG100-1	106.5^3	1820^3	2×1820^3	1820^3
TNG300-1	300^3	2500^3	1×2500^3	2500^3

Table 2.3: The volume and quantity of gas, tracer, and dark matter particles within the simulations with the highest resolution realisations, as given by Nelson et al. [21].

box, TNG300, facilitates the study of galaxy clustering and allows for the analysis of rare and massive objects, such as galaxy clusters while providing the largest statistical sample of galaxies. Conversely, the smaller physical volume of TNG50 offers over a hundred times better mass resolution than TNG300, enabling a more detailed examination of the structural properties of galaxies and small-scale gas phenomena in and around them. The TNG100 simulation, situated between the two extremes, utilises the same initial conditions as the original Illustris simulation but is adjusted for the updated cosmology. This design enables robust comparisons between the results from the original Illustris and the updated TNG model.

Throughout these simulations, a Monte Carlo tracer particle scheme [31] is employed to track the Lagrangian evolution of baryonic matter, according to Nelson et al. [21]. An on-the-fly cosmic shock finder, coupled to the code [32, 33], is utilised. For comprehensive descriptions of the behaviour, implementation, parameter selection, and validation of these physical models, refer to the methods papers by Weinberger et al. [30] and Pillepich et al. [26]. The updated initial conditions are provided by Nelson et al. [21] and are illustrated in table 2.2.

Varying resolutions are available for the same volumes in the TNG simulations: high resolution (indicated by the -1 suffix, e.g., TNG100-1), intermediate resolution (indicated by the -2 suffix), and low resolution (indicated by the -3 suffix). Each resolution level differs by a factor of two in spatial resolution and eight in mass resolution. For detailed information regarding the volumes and the quantities of gas, tracer, and dark matter resolution elements, please refer to table 2.3.

2.3 X-Ray Imaging

According to Teubner & Brückner [34], different carrier waves, imaging systems, and imaging methods are employed in various contexts. For instance, acoustical waves are used for ultrasonic testing in medical and material sciences, microwaves serve in radar imaging, and X-rays and particle beams, such as electrons, are utilised for imaging small objects. The origins of X-ray astronomy can be traced back to the immediate aftermath of World War II, as noted by Arnaud et al. [35]. During this period, surplus military rockets were repurposed to carry radiation detectors above the Earth's atmosphere for short durations, primarily focusing on studying X-ray emissions from the solar corona.

Arnaud et al. [35] state that a significant advancement in X-ray astronomy technology was the adoption of X-ray-sensitive CCDs, which offered superior imaging and spectroscopic capabilities compared to the previously used imaging proportional counters or scintillators. The launch of ASCA in 1993 marked the first deployment of CCDs in an X-ray astronomy

satellite. In this field, prominent observatories such as NASA’s Chandra, ESA’s X-ray Multi Mirror (XMM)-Newton, and JAXA’s Suzaku are operational. Chandra’s primary strength lies in its high-resolution telescopic capabilities, enabling the capture of detailed images and spectral data through the use of gratings. In contrast, XMM-Newton boasts the largest focusing area among X-ray telescopes, while Suzaku covers a broad energy bandpass and maintains minimal background interference.

Unlike other wavebands, X-ray detection typically occurs on an event-by-event basis rather than as a bolometric flux over a specific frequency range, as highlighted by Arnaud et al. [35]. The value of high-resolution imaging in X-ray astronomy becomes evident through several advantages. These include the ability to resolve sources with minimal angular separation, delineate different regions within a single source, utilise image morphology to inform quantitative data modelling, and effectively gather and concentrate photons due to the low flux of X-ray sources, with the weakest sources emitting less than one photon per day. Moreover, high-resolution imaging enables the simultaneous measurement of sources of interest and the determination of contaminating background radiation using other detector regions.

Arnaud et al. [35] note that dispersive spectrometers, such as transmission or reflection gratings, further enhance observational capabilities by providing high spectral resolution, thereby allowing for more detailed analyses of the X-ray emissions from celestial objects. This combination of high-resolution imaging and advanced spectroscopic techniques significantly improves the understanding of the physical processes occurring in X-ray-emitting sources.

Teubner & Brückner [34] introduce the term flux in the context of electromagnetic radiation, particularly light. They explain that electromagnetic radiation can be characterised by waves or by considering energetic particles travelling through space, known as photons. This particle perspective allows for an analogy with electric current, where the current’s magnitude is expressed as particles per unit time, specifically, the number of charges per unit time in the case of electricity. This analogy leads to the concept of photon current, which represents the number of photons emitted per unit time. This is referred to as *radiant flux* Φ_e . Radiant flux is equivalent to the energy emitted over a given time interval, making it synonymous with optical power, measured in watts.

2.3.1 X-Ray Detectors

Detectors play a vital role in X-ray astronomy by capturing and converting incoming X-rays into electronic signals, which are then transformed into data for analysis, as described by Arnaud et al. [35]. The primary function of these detectors is to extract detailed information about the incident X-rays, including their energy, arrival time, quantity, detection location (ideally in two dimensions), and polarisation state. X-ray detectors also often detect other forms of ionising radiation, and in some cases, even optical or infrared light, making it crucial to distinguish X-ray photons from particles like energetic charged particles.

Arnaud et al. [35] describe the evolution of different types of X-ray detectors, each offering specific advantages and disadvantages. Historically, *proportional counters* were used, consisting of a gas-filled chamber with a high-voltage wire and an X-ray window. X-rays ionise the gas, causing electron avalanches and creating a charge pulse. However, this method becomes less effective for energies beyond about 50 keV. Another option are *scintillators*, which use crystals that emit optical light when they absorb X-ray photons, providing a way to detect higher-energy X-rays. For applications requiring high spatial resolution but less emphasis on energy resolution, *Microchannel Plate (MCP)* are a good choice. They comprise numerous small pores or tubes arranged in an electric field. When an X-ray produces

a photoelectron, an avalanche of electrons occurs within the pore, providing precise spatial resolution but sacrificing spectral information. *Cadmium Zinc Telluride (CZT)* detectors, now commonly used for X-rays with energies above 10 keV, operate similarly to CCDs but offer superior energy resolution in the 10 keV to 100 keV range due to their higher quantum efficiency. Unlike CCDs, CZT detectors must read each pixel individually, as there is no charge transfer across rows and columns. With energy resolutions of a few per cent achievable in the 10 keV to 100 keV energy range, CZT is preferred over CCD at these energies due to its better Quantum Efficiency (QE). To address limitations in X-ray spectroscopy using gratings on telescopes like Chandra and XMM-Newton, quantum *microcalorimeter* technology was developed. These devices measure temperature changes with high precision, offering excellent energy resolution and efficiency, especially for extended sources. However, they require complex cryogenic systems to operate at temperatures in the mK range.

2.3.2 Charge-Coupled Devices

In the late 1970s, a significant breakthrough in imaging technology occurred with the introduction of the Charge-Coupled Device (CCD), developed at Bell Laboratories in 1969 [36]. As highlighted by Arnaud et al. [35], CCDs quickly surpassed earlier technologies and revolutionised astronomy in the 1980s due to their exceptional sensitivity and linear brightness response. Today, they are the most widely used detectors in optical astronomy and are integral to nearly all commercial digital cameras.

A CCD is an array of interconnected capacitors, which can be either a MOS structure or a p - n junction. Arnaud et al. [35] explain that a basic *Metal-Oxide-Semiconductor (MOS)* structure consists of a metal gate separated from a p -type semiconductor by an insulator. However, this design suffers from charge trapping at the semiconductor-insulator interface, which hinders the performance by preventing the easy transfer of electrons to the readout. A layer of n -type semiconductor is added between the p -type material and the oxide to mitigate this issue, creating what is known as a buried channel CCD. In contrast, a p - n junction is formed by contacting p -type and n -type materials, according to Chromey [37]. The fully depleted p - n junction CCDs, as described by Arnaud et al. [35], have thicker active regions compared to standard MOS CCDs, making them more efficient at detecting high-energy X-rays. The different types of semiconductors are created by doping them with different impurity atoms. As an example, Arnaud et al. [35] state that doping with phosphorus creates an n -type semiconductor with excess free electrons, and doping with boron produces a p -type semiconductor with excess free holes. Layering differently doped material can create structures that encourage or discourage electric current.

Arnaud et al. [35] explain that the charge collection and transfer process within CCDs is generally similar despite their different physical structures. Photons interacting with the semiconductor substrate (usually silicon) generate electron-hole pairs. An electric field is applied to collect the electrons, which are stored in pixels during a period known as frame time. Each pixel is connected to its neighbours, allowing the accumulated charge to be transferred pixel by pixel to the readout amplifier. Sequentially applying higher voltage to adjacent gates facilitates this transfer, with a brief overlap in time allowing the charge to move efficiently. As Arnaud et al. [35] explain, the term Charge-Coupled Device refers to this temporary joining of adjacent potential wells to transfer the charge. Once in the readout amplifier, the charge is converted into digital data for analysis. For optimal X-ray sensitivity, the ideal energy range is typically from a few hundred eV to 10 keV. Photons with energies below a few hundred eV may not be fully detected if they interact close to the surface. In contrast, photons with high energies are more likely to pass through without interacting, according to Arnaud et al. [35].

For X-ray applications, CCDs are typically operated in photon-counting mode, which differs from the continuous accumulation used in optical applications, as stated by Arnaud et al. [35]. This mode allows for precisely determining each X-ray photon's position and energy. In optical applications, individual photons produce too little signal to be detected independently, requiring longer frame times to accumulate thousands of photoelectrons.

Arnaud et al. [35] note two main approaches to improving the detection efficiency of low-energy X-rays in CCDs for astrophysical purposes. Firstly, thinning or creating „open“ gates in the CCD by etching holes, therefore reducing the absorption and allowing more photons to reach the active region of the detector. Secondly, back-illuminating the CCD. In this configuration, the CCD is flipped so that the gate structures are on the side away from the X-ray source, while the back layer facing the X-rays is thinned. The thinning reduces the thickness of dead layers, significantly improving low-energy QE compared to front-illuminated CCDs. However, back-illuminated CCDs are more prone to noise and charge-transfer inefficiency due to surface defects, leading to lower resolution.

2.4 The Satellite XMM-Newton

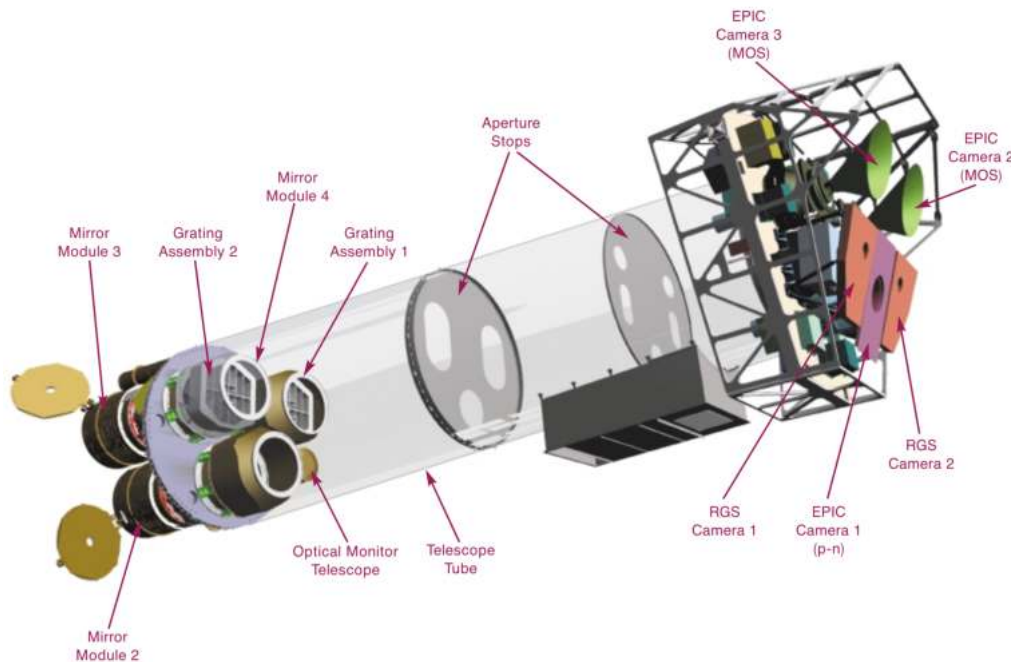


Figure 2.1: Schematic overview by Bagnasco et al. [38] of the satellite and its components.

The XMM-Newton spacecraft is equipped with three advanced X-ray telescopes, two of which carry European Photon Imaging Camera (EPIC)-MOS cameras and Reflection Grating Spectrometer (RGS), while the third features the specially designed EPIC-pn camera for enhanced X-ray imaging capabilities, as described by Jansen et al. [39]. Turner et al. [40] further elaborate that the RGSs divert about 50 % of the X-ray flux from the EPIC beams. Taking into account structural obscuration, approximately 44 % of the original flux reaches the MOS cameras, whereas the EPIC-pn camera benefits from an unobstructed X-ray beam. Furthermore, all three cameras share a common forward section that houses a variety of essential components: a filter wheel, door, calibration source, radiation shielding, the spacecraft's focal plane bulkhead interface, and the internal bulkhead forming part of the camera's vacuum enclosure. This shared structure can be observed in fig. 2.2a, while a schematic overview of the entire satellite is presented in fig. 2.1.

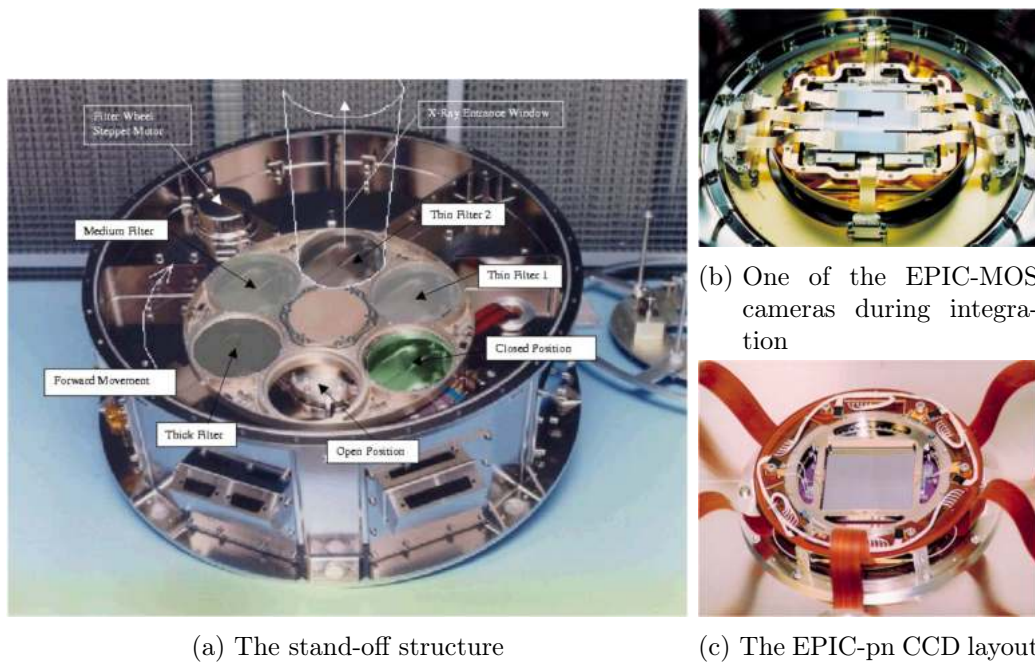


Figure 2.2: Different components of the XMM-Newton satellite as shown in Bagnasco et al. [38].

According to the XMM-Newton User Handbook (UHB) [41], there is also a smaller 30 cm optical/UV telescope with a so-called Optical Monitor (OM) in addition to the X-ray telescopes allowing simultaneous access to both the X-ray and the optical/UV spectrum. These six scientific instruments constitute the entirety of the spacecraft's instrument payload. These can operate simultaneously unless precluded by predefined constraints or independently and in various data acquisition modes. The telescopes have an effective collection area of 4500 cm^2 at 1 keV and 1000 cm^2 at 10 keV [42] with a total collecting mirror area of 120 m^2 [43]. According to Arnaud et al. [35], the effective area is the largest of a focusing telescope ever.

2.4.1 The EPIC-MOS Cameras

Turner et al. [40] provide a detailed description of the EPIC-MOS cameras and their telescopes. Each camera's focal plane has seven CCDs, where the central CCD is positioned directly at the focal point along the telescope's optical axis. The other six CCDs are stepped 4.5 mm closer to the mirrors to account for the curved nature of the focal plane. This arrangement improves focus on off-axis sources and minimises dead space by overlapping adjacent CCDs by 1 mm, creating a $300 \mu\text{m}$ overlap, as explained by Turner et al. [40]. The layout of the CCDs in the focal plane is depicted in fig. 2.2b. The overall imaging area of each EPIC-MOS camera covers roughly $2.5 \text{ cm} \times 2.5 \text{ cm}$, with the mosaic of seven CCDs spanning a diameter of 62 mm across the focal plane. Each individual CCD consists of 600×600 , $40 \mu\text{m} \times 40 \mu\text{m}$ pixels, where each pixel corresponds to $1.1'' \times 1.1''$ of the Field of View (FOV). The two MOS cameras are mounted onto the focal plane bulkhead at perpendicular angles, with a 90° rotation between them. This configuration ensures that the outer CCDs of one camera cover the gaps left by the other camera, as detailed by Turner et al. [40]. Additionally, the spaces around the central CCD overlap, preventing the detection of photons passing through these gaps. Figures 2.3a and 2.3b illustrate the numbering scheme and the detector coordinates for EPIC-MOS1 and EPIC-MOS2, respectively.

According to the UHB [41], some of the CCDs in the XMM-Newton's EPIC-MOS cameras

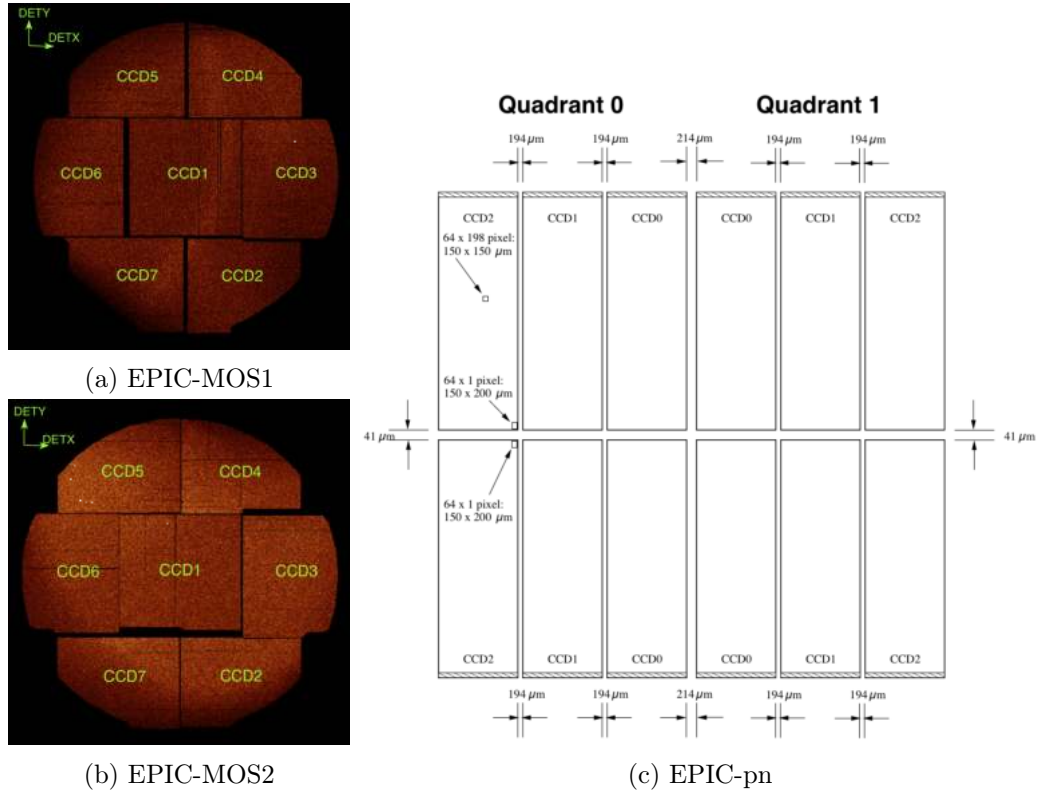


Figure 2.3: Numbering scheme of the EPIC-MOS cameras as given in the UHB [41] and the internal boundaries of EPIC-pn as given by Strüder et al. [44]. Note that EPIC-MOS1 is rotated by 90° on the satellite.

are no longer fully operational. On March 09, 2005, CCD 6 of EPIC-MOS1 began returning signals at saturation levels, suggesting possible severe damage. As a result, CCD 6 was turned off for subsequent observations. Similarly, on December 11, 2012, CCD 3 of EPIC-MOS1 also sustained significant damage, leading to its deactivation.

2.4.2 The EPIC-pn Camera

Strüder et al. [44] describe the EPIC-pn camera and its telescope. The telescope achieves an angular resolution of $15''$ Half Energy Width (HEW) at both 1.5 keV and 8 keV, requiring a position resolution of $540 \mu\text{m}$ at the focal plane. The Full Width Half Maximum (FWHM) of the Point Spread Function (PSF) is $6.6''$, and to match an angular resolution capability of $3.4''$, a pixel size of $150 \mu\text{m} \times 150 \mu\text{m}$ ($4.1'' \times 4.1''$) was selected, along with a position resolution of $120 \mu\text{m}$. To ensure high radiation hardness, active MOS structures were avoided, and fast charge transfer to a depth of over $10 \mu\text{m}$ beneath the surface was implemented. Figure 2.2c shows the final layout of the EPIC-pn detector, which consists of four quadrants. Each quadrant contains three pn-CCD subunits, each formatted as 200×64 pixels, operated in parallel. Uniform detector quality across the FOV is achieved by fabricating the twelve pn-CCDs, each $3 \text{ cm} \times 1 \text{ cm}$, monolithically on a single wafer. Figure 2.3c illustrates the insensitive or partially sensitive gaps between the CCDs and quadrants. Since all CCDs are integrated on the same wafer, the relative alignment of chips and pixels is precise to within $1 \mu\text{m}$. The division of the focal plane into subunits adds redundancy, and CCD0 in quadrant 1, where the focal point lies, covers approximately 97% of the telescope's FOV. An additional 6 cm^2 of the CCD sensitive area extends beyond the FOV and is utilised for background studies.

2.4.3 Background Component of the EPIC Cameras

The XMM-Newton EPIC Background Working Group was established in 2005, as noted by Carter & Read [45], to improve the understanding of the EPIC cameras' image background. This understanding is crucial for accurately analysing data and addressing challenges such as SR and DN. However, the group ceased its operations in 2012, according to ESA's website [46].

Carter & Read [45] describe the EPIC background as consisting of three main components: photons, particles, and electronic noise. The *photon*-related background is divided into hard and soft X-ray emissions. Hard X-ray photons originate primarily from unresolved Active Galactic Nuclei (AGN) within the FOV, as well as from reflections of out-of-FOV sources into the FOV. These out-of-FOV sources contribute around 7 % of the total in-FOV flux, remaining relatively constant between observations and following a predictable power-law distribution. On the other hand, soft X-ray photons come from multiple sources, including the Local Bubble, the Galactic Disk, the Galactic Halo, and Solar Wind Charge Exchange (SWCX). These soft X-ray sources exhibit slight variability, with changes of up to 35 % between observations, particularly during long observation periods when SWCX may play a role. Unlike the hard X-ray background, which is spatially vignettted like all X-rays, the soft X-ray background is much more complex and comprises numerous components. Soft and hard X-ray photons are spatially vignettted, as with all X-ray photons, and the only structure seen is from real astronomical objects. The *particle* background includes soft protons and instrumental background. Soft protons, originating from the Sun, can have a significant observational impact, ranging from 30 % to 40 %, and are especially unpredictable during solar flares. Their frequency increases closer to perigee due to the Earth's magnetosphere trapping them. They vary in spectral intensity and shape, are observed only inside the FOV and are similarly distributed over the detector to X-rays but with a flatter vignetting function than for photons. No spatial structure can be identified in the EPIC-pn detector, but some structure may occur due to the RGS in front of the EPIC-MOS detectors. Additionally, high-energy particles contribute to the instrumental background through direct charge generation or interaction with the detector, causing variations of up to 10 % within observation and sometimes up to tenfold across observations, particularly during solar flare activity. This background component is spatially distributed differently from X-rays and is not vignettted. The final component, *electronic noise*, is influenced by factors such as bad pixels, bright columns, and readout noise in the EPIC-pn detector. While dark current is considered negligible, bright pixels and columns can fluctuate by as much as 10 % between observations. Readout noise in the EPIC-pn detector varies depending on the mode, with the extended-full-frame mode being the least affected and the small window mode the most impacted. Low-energy enhancements in the outer CCDs of the EPIC-MOS detectors, which are more pronounced during high background rates, can also influence the noise profile, potentially affecting up to 20 % of observations. The electronic noise is distributed differently than genuine X-ray photons and is not vignettted, apart from the low-energy enhancement in the MOS detectors. Bright pixels and columns are seen at specific locations, and structure is seen near the EPIC-pn readout. A summary and visualisations are available in Read and Ponman [47].

2.5 Simulating X-Ray Satellites

Various simulation tools have been developed for specific instruments in X-ray astronomy, as highlighted by Dauser et al. [48]. Notable examples include *marx* [49] for Chandra, *SciSim* [50] for XMM-Newton, and *NuSim* [51, 52] for NuSTAR. However, these tools are designed for individual missions. In contrast, *simx* [53] allows for rapid simulation of observations across multiple X-ray missions. However, it has limitations regarding detailed

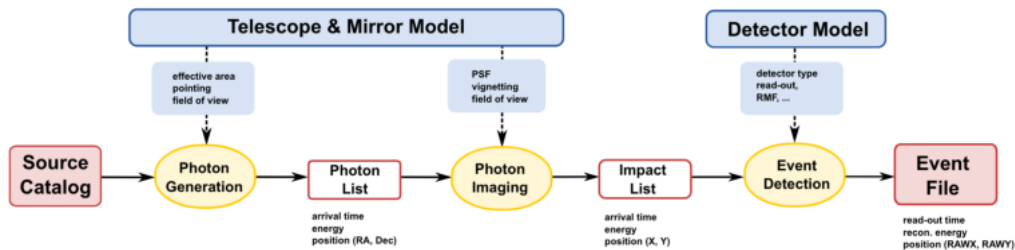


Figure 2.4: Flow chart by Dauser et al. [48] illustrating the three major functional blocks of SIXTE.

timing investigations and only offers a narrow selection of detector types and effects, according to Dauser et al. [48]. While *simx* is useful for gaining a broad overview of an instrument’s capabilities, it is less suited for conducting comprehensive studies of scientific performance.

To meet the demand for detailed modelling of detector effects for complex astrophysical sources, Dauser et al. [48] introduced the *Simulation of X-ray Telescopes (SIXTE)* software. Like other simulators such as *marx* and *SciSim*, SIXTE is designed to deliver realistic and representative simulations of future instruments for complex astrophysical scenarios. Additionally, it supports instrument development by offering detailed modelling of detector characteristics. What distinguishes SIXTE from other simulators is its ability to incorporate new instruments seamlessly without requiring alterations to the software or an in-depth understanding of its internal workings. Its modular design also allows for easy replacement of components with more sophisticated and detailed descriptions. Unlike a complete simulator like *marx*, SIXTE does not predict the behaviour of optics but instead relies on calibration data, such as vignetting, PSF, Ancillary Response File (ARF), and Redistribution Matrix File (RMF), as input. The last three terms are explained later on. This method significantly reduces the computational burden of simulations while maintaining flexibility, allowing using simulated or measured calibration data of varying complexity to describe the instrument, as noted by Dauser et al. [48].

Dauser et al. [48] explain that the SIXTE simulator employs a Monte Carlo framework that simulates astrophysical phenomena by tracking individual photons. Before any simulation can begin, input files containing data on astrophysical sources are collected in a Flexible Image Transport System (FITS)-based format known as *Simulation Input (SIMPUT)* [54]. This format provides a source catalogue detailing the instrument-independent properties of one or more X-ray sources, such as their positions and observed flux within a specific energy band. This information is essential for generating a photon sample with key characteristics, including time, energy, and direction of origin, as described by Dauser et al. [48].

The simulation process, outlined by Dauser et al. [48], begins with photon imaging. During this stage, photons are passed through the optics and projected onto the detector plane, producing photon impacts (or events) that are recorded with specific times, energies, and positions. The resulting event file captures this data for all photon impacts. In the current version of SIXTE, pre-computed or measured calibration files are used to simulate the imaging process, taking into account factors such as the detection probability for off-axis versus on-axis photons, the vignetting function, and the PSF. Additionally, SIXTE allows users to define the detector by configuring a rectangular array of pixels, with dimensions specified in an Extensible Markup Language (XML) file. Each pixel is uniquely identified by its **RAWX** and **RAWY** coordinates, and in the case of a CCD, the charge is shifted to the readout nodes at the detector’s edge. Figure 2.4 provides an illustration of SIXTE’s functionality. Moreover, SIXTE is the official end-to-end instrument simulator for *eROSITA* aboard the *Spektrum-Roentgen-Gamma* and is also planned for use with the *Athena* satellite for both

the Wide Field Imager and the X-ray Integral Field Unit instruments.

The image of a distant point source of light is never truly a point. Instead, it is described by a PSF, which demonstrates how light from a point source, like a star, is dispersed across multiple pixels in the image, as outlined by McLean et al. [55]. The PSF can change across the FOV, with its shape influenced by various factors affecting image quality. As Arnaud et al. [35] and George et al. [56] explain, the ARF represents the detector system’s „effective area,“ typically including elements such as mirror vignetting and filters. In addition, the RMF defines the likelihood of detecting a particular signal for a given input photon energy..

2.6 Artificial Intelligence

According to Goodfellow et al. [57], although deep learning has recently surged in popularity, it has a rich and varied history, spanning different philosophical approaches and alternating periods of prominence. Contrary to its perceived novelty, deep learning remained relatively obscure for many years and underwent several name changes before being labelled as *deep learning*. Goodfellow et al. [57] explain that early learning algorithms were initially conceived as computational models of biological learning, aiming to mimic the brain’s processes, which led to these models being called artificial neural networks.

The origins of deep learning can be traced back to the 1940s, according to Goodfellow et al. [57]. The first wave of development began with theories on biological learning [58, 59], which fostered the rise of the term *cybernetics* and the creation of models that attempted to simulate a single neuron, such as the perceptron introduced by Rosenblatt [60, 61]. However, the limitations of linear models like the perceptron, particularly its inability to learn the XOR function, caused a decline in the popularity of neural networks during the 1960s.

The introduction of the backpropagation algorithm by Rumelhart et al. [62] in 1986 marked a significant resurgence of neural networks, as noted by Goodfellow et al. [57]. This algorithm enabled the training of networks with one or two hidden layers, reinforcing the idea that intelligent behaviour could emerge from networks of simple computational units. During this period, the term *connectionism* was used to describe neural networks due to the layered structure and interconnected neurons. However, despite the initial excitement, exaggerated claims and subsequent failures to meet expectations led to neural networks falling out of favour during the 1990s.

The revival of neural networks occurred in 2006 with the introduction of greedy layer-wise pretraining by Hinton et al. [63], particularly for deep belief networks. As described by Goodfellow et al. [57], this technique enabled deeper neural networks to be trained more effectively, thus popularising the term *deep learning* and marking the beginning of its modern resurgence.

2.6.1 Perceptrons

According to Goodfellow et al. [57], simple linear models such as the *perceptron*, introduced by Rosenblatt [60, 61], laid the groundwork for modern deep learning. The perceptron connects a set of input values $\mathbf{x} \in \mathbb{R}^n$ with an output y by learning a set of weights $\mathbf{w} \in \mathbb{R}^n$, as expressed in eq. (2.2).

$$f(\mathbf{x}, \mathbf{w}) = \sum_{i=1}^n x_i w_i \quad (2.2)$$

However, this simple model could only solve linearly separable problems, such as AND and OR, but not more complex functions like XOR.

To address this limitation, Goodfellow et al. [57] highlight the development of *Multilayer Perceptrons (MLPs)* and methods for calculating the gradient, allowing the network to learn non-linear functions. Calculating the gradient and the learning process are introduced in section 2.6.6. MLPs are neural networks with one or more hidden layers between the input and output layers. As Goodfellow et al. [57] explain, their goal is to approximate a target function f^* . The network defines a mapping $\mathbf{y} = f(\mathbf{x}; \boldsymbol{\theta})$, where the parameters $\boldsymbol{\theta}$ include the weights $\mathbf{w}$ associated with the inputs and results between hidden layers and a bias b , which shifts the input space. These networks are often called *feed-forward* because information flows in one direction: from input $\mathbf{x}$, through intermediate functions f , to the output $\mathbf{y}$.

MLPs are also termed *networks* because the composition of various functions represents the network, which is associated with a directed acyclic graph. Goodfellow et al. [57] illustrate this by explaining that a chain of three functions f_1 , f_2 , and f_3 can be combined as shown in eq. (2.3).

$$f(\mathbf{x}) = f_3(f_2(f_1(\mathbf{x}))) \quad (2.3)$$

This structure, where functions are composed in layers, gives rise to the term *deep learning*. The depth of the model corresponds to the number of layers, and the last layer is the output layer. Intermediate layers, referred to as *hidden layers*, are crucial to the network's ability to approximate f^* .

The term *neural* is used to describe these networks due to their loose inspiration from neuroscience, according to Goodfellow et al. [57]. Each neuron computes an activation based on the input from other units and a non-linear activation function, as Russell & Norvig [64] emphasise. Non-linearity is crucial because if the activation function were linear, the network would represent only a linear function, even with multiple layers. Instead, non-linear activations allow the network to model complex, non-linear relationships. The common types of activation functions are illustrated in fig. 2.5.

By incorporating non-linearities in its nodes, the MLP becomes a universal approximator, as Du & Swamy [65] explain. It has been proven that a three-layer MLP with a sigmoid activation function can approximate any continuous multivariate function to arbitrary precision, highlighting the power and flexibility of deep learning models. The eq. (2.4), where $h_i(\mathbf{x}) = g_i(f_i(\mathbf{x}))$ and g_i is the activation function for f_i , gives an idea of the typical structure of neural networks used in deep learning today, as explained by Russell & Norvig [64].

$$f(\mathbf{x}) = h_3(h_2(h_1(\mathbf{x}))) \quad (2.4)$$

2.6.2 Convolutional Neural Networks

According to Russell & Norvig [64], the introduction of CNNs by LeCun et al. [66] was a significant advancement in overcoming the limitations of fully connected networks for image tasks. CNNs are characterised by spatially local connections and weight sharing across layers, allowing them to handle large images and efficiently learn spatial hierarchies of features. The convolution operation at the core of CNNs applies kernels (small, fixed-size filters) to input data. For a one-dimensional input vector $\mathbf{x} \in \mathbb{N}^n$, representing image pixels, and a kernel $\mathbf{k} \in \mathbb{R}^l$, the convolution operation $\mathbf{z} = \mathbf{x} * \mathbf{k}$ is given in eq. (2.5) as described by Russell & Norvig [64].

$$z_i = \sum_{j=1}^l k_j x_{j+i(l+1)/2} \quad (2.5)$$

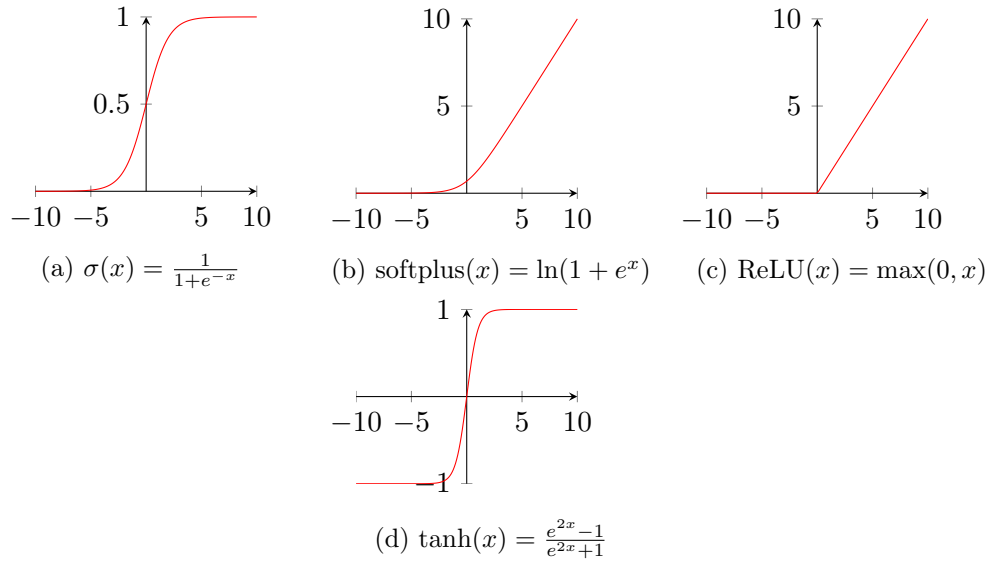


Figure 2.5: The most commonly used activation functions: (a) Logistic (or sigmoid) function, (b) softplus (a smoother version of Rectified Linear Unit (ReLU)), (c) ReLU, and (d) hyperbolic tangent.

Russell & Norvig [64] further explain that this operation calculates the dot product between the kernel and a segment of the input, capturing local patterns in the data. By applying the kernel with a stride greater than one, CNNs can reduce the number of output pixels while preserving crucial information, often aided by padding techniques.

Russell & Norvig [64] explain that an image cannot simply be treated as a vector of pixel values because the relationships between neighbouring pixels carry essential information about the structure within the image. If a fully connected neural network used pixel values as input, it would yield the same results whether the image remained unaltered or its pixels were shuffled, as the network would ignore the spatial structure. Furthermore, for an image with n pixels and n neurons in the first hidden layer, a fully connected network would require n^2 weights. This would create a vast parameter space, necessitating extensive training data and computational power to train effectively.

To address these issues, CNNs are designed, according to Russel & Norvig [64], so that each neuron in the first hidden layer only receives input from a small, localised region of the image. This design has two significant benefits. Firstly, it preserves the spatial relationships between pixels by focusing on local adjacency. Secondly, it drastically reduces the number of parameters. If each local region consists of $l \ll n$ pixels, the total number of weights required becomes $ln \ll n^2$, shrinking the parameter space and thus improving efficiency.

Additionally, as Russell & Norvig [64] note, images exhibit approximate spatial invariance, where similar features often appear in multiple image regions. To take advantage of this, CNNs implement weight sharing across neurons, meaning the weights used to connect one local region to a hidden unit are applied across all regions. This way, the hidden units act as feature detectors, recognising the same patterns irrespective of their position in the image. To detect multiple features, CNNs usually employ several detectors for each local region, each with its own set of weights. As a result, the number of parameters is reduced to $ld \ll n^2$, where d is the number of distinct feature detectors.

According to Goodfellow et al. [57], CNNs excel in three critical areas that significantly enhance the performance of machine learning systems. First, they use *sparse interactions*, where small kernels (filters) reduce the number of parameters compared to traditional networks, leading to lower memory usage and faster computation. This efficiency allows

CNNs to learn effectively, even with limited data. Second, CNNs employ *parameter sharing*, meaning that the same set of kernel weights is applied across the entire input, unlike standard neural networks where each weight is used only once per connection. This approach reduces the number of parameters and improves the network’s ability to generalise across various input regions. Finally, CNNs demonstrate *equivariance to translation*, meaning that when an input shifts, the output shifts correspondingly. This property is crucial for tasks such as object detection, where feature positions may vary within an image.

2.6.3 Generative Adversarial Networks

According to Goodfellow et al. [57], GANs [67] are built on a game-theoretic framework where two neural networks, the generator and the discriminator, compete against each other. The generator network is responsible for creating synthetic samples, which it produces directly as $\mathbf{x} = g(\mathbf{z}; \boldsymbol{\theta}_g)$, where $\mathbf{z}$ is a noise vector sampled from a prior distribution, and $\boldsymbol{\theta}_g$ represents the parameters of the generator. On the other hand, the discriminator network is tasked with distinguishing between actual samples from the training dataset and the synthetic samples generated by the generator. It outputs a probability value denoted by $d(\mathbf{x}; \boldsymbol{\theta}_d)$, indicating the likelihood that $\mathbf{x}$ is an authentic training example rather than a synthetic sample created by the model.

The training process of GANs, as explained by Goodfellow et al. [57], involves both networks simultaneously improving their performance through adversarial training. The generator strives to produce increasingly convincing samples for the discriminator while the discriminator works to become more adept at identifying real versus fake samples. This dynamic creates a minimax game where the generator aims to maximise the probability of the discriminator making a mistake, and the discriminator seeks to minimise its error rate. As the training progresses, both networks ideally converge, resulting in a generator capable of producing highly realistic samples that closely resemble the original training data.

One of the key motivations behind the design of GANs, according to Goodfellow et al. [57], is that their learning process does not require approximate inference or the computation of gradients for a partition function. When $\max_d v(g, d)$, with v being the payoff of the discriminator, is convex in $\boldsymbol{\theta}_g$, the method is assured to converge and is asymptotically consistent. However, challenges arise in learning within GANs when $\max_d v(g, d)$ is not convex, mainly when both g and d are implemented as neural networks. In general, simultaneous gradient descent on the costs of both players does not guarantee the attainment of equilibrium. Empirical evidence suggests that in the most effective formulations of the GAN game, the generator’s goal is to maximise the log probability that the discriminator makes an error rather than merely minimising the log probability of correct predictions by the discriminator.

2.6.4 Transformers

The groundbreaking paper by Vaswani et al. [68] introduced the transformer architecture, which employs a self-attention mechanism to model long-distance context without relying on sequential dependencies. According to Russell & Norvig [64], traditional sequence-to-sequence models used attention directed from the target Recurrent Neural Network (RNN) to the source RNN, with Recurrent Neural Networks allowing cycles, unlike feed-forward networks. Self-attention extends this idea by allowing each sequence of hidden states to attend to itself, source to source and target to target, thereby capturing long-distance and nearby contexts.

Russell & Norvig [64] explain that the most straightforward way to implement self-attention is by constructing an attention matrix using the dot product of input vectors. However, this

has a drawback: the dot product between a vector and itself is always high, leading to a bias where each hidden state disproportionately attends to itself. The transformer overcomes this by projecting the input into three distinct representations through separate weight matrices. First, the *query vector* $\mathbf{q}_i = \mathbf{W}_q \mathbf{x}_i$ represents the vector being attended from, akin to the target in standard attention mechanisms. Second, the *key vector* $\mathbf{k}_i = \mathbf{W}_k \mathbf{x}_i$ represents the vector being attended to, analogous to the source. Finally, the *value vector* $\mathbf{v}_i = \mathbf{W}_v \mathbf{x}_i$ provides the contextual information.

Self-attention is just one component of the transformer model. As described by Russell & Norvig [64], each transformer layer contains multiple sub-layers. The self-attention operation is followed by feed-forward layers, where the same weight matrices are applied independently at each position. A non-linear activation function, typically ReLU, is used after the first feed-forward layer.

According to Lepcha et al. [69], large versions of transformer models are commonly trained and then fine-tuned for specific tasks, making them effective even with limited data. Additionally, the performance of these models tends not to plateau as their size increases. Larger models yield more accurate results and often reveal fascinating emergent properties.

2.6.5 Autoencoders

Many unsupervised deep learning algorithms are rooted in the concept of an autoencoder, as described by Goodfellow et al. [57]. An autoencoder consists of two main components: an encoder and a decoder. The encoder maps the input data $\mathbf{x}$ to a latent representation $\mathbf{z}$, often referred to as an embedding. The decoder then maps this representation $\mathbf{z}$ back to the original input $\mathbf{x}$. The encoder is typically a parameterised function denoted as f , while the decoder is represented as g . The goal of training the autoencoder is to ensure that the output closely approximates the original input, i.e., $\mathbf{x} \approx g(f(\mathbf{x}))$, implying that the decoding process effectively reverses the encoding.

According to Goodfellow et al. [57], the functions f and g can take many forms, from simple linear models parameterised by a single matrix to more complex representations using deep neural networks. This flexibility allows autoencoders to capture intricate patterns in data, making them a powerful tool in various unsupervised learning tasks.

Goodfellow et al. [57] highlight that autoencoders are unsuitable if they replicate their input perfectly. Instead, they are typically designed to only approximately copy the input, with restrictions encouraging the model to focus on reproducing input that resembles the training data. It can learn meaningful data properties by forcing the model to prioritise certain input aspects.

The aim is not merely to replicate the input exactly but to train the model to perform the copying task in a way that yields useful representations, as explained by Goodfellow et al. [57]. This is achieved through the latent representation $\mathbf{z}$, which can capture valuable data characteristics. A standard method to extract these features is to restrict the dimensionality of $\mathbf{z}$ to be smaller than that of the input $\mathbf{x}$. This type of autoencoder, known as an undercomplete autoencoder, is designed to force the model to compress the input and, in doing so, learn the most critical features of the data.

Goodfellow et al. [57] highlight that training an autoencoder involves balancing two crucial factors. The first is learning an embedding, $\mathbf{z}$, such that the input $\mathbf{x}$ can be approximately reconstructed from $\mathbf{z}$ through a decoder. This process is grounded in the assumption that $\mathbf{x}$ is drawn from the training data, meaning the autoencoder only needs to reconstruct examples typical of the data-generating distribution accurately. The second factor is imposing a constraint or regularisation penalty, which could be an architectural constraint limiting the model's capacity or adding a regularisation term to the reconstruction cost.

These techniques favour solutions that make the model less sensitive to input variations, encouraging the autoencoder to focus on the most meaningful aspects of the data.

As Goodfellow et al. [67] explain, neither factor alone is beneficial. Copying the input perfectly is not informative, and ignoring the input is equally unhelpful. However, when combined, these forces guide the model toward capturing the structure of the data-generating distribution. Furthermore, the core principle is that the autoencoder must only represent the variations necessary to reconstruct examples from the training data. If the data distribution is concentrated around a low-dimensional manifold, this results in representations that implicitly encode a local coordinate system for the manifold. The autoencoder's encoder learns to map the input space $\mathbf{x}$ to a representation space $\mathbf{z}$ that is sensitive only to variations along the manifold's directions and insensitive to changes orthogonal to it.

A simple form of an autoencoder, as described by Russel & Norvig [64], is the linear autoencoder, where both the encoder f and decoder g are linear functions sharing a weight matrix $\mathbf{W}$. This is illustrated in eqs. (2.6) and (2.7).

$$\mathbf{z} = f(\mathbf{x}) = \mathbf{W}\mathbf{x} \quad (2.6)$$

$$\mathbf{x} = g(\mathbf{z}) = \mathbf{W}^T \mathbf{z} \quad (2.7)$$

The aim is to adjust the weight matrix $\mathbf{W}$ so that the low-dimensional latent representation $\mathbf{z}$ retains sufficient information for reconstructing the original high-dimensional input $\mathbf{x}$. This linear autoencoder primarily captures basic structures but is limited in modelling complex data relationships.

The variational autoencoder provides a probabilistic framework for capturing more intricate data patterns. For further details, Russel & Norvig [64] offer an explanation, and the original work by Kingma and Welling [70] lays the foundational theory.

2.6.6 The Learning Process

When training a neural network, the primary task is optimisation, according to Goodfellow et al. [57], which involves adjusting the input x to either minimise or maximise a function $f(x) = y$. Most optimisation problems are framed as minimisation tasks; if the objective is to maximise $f(x)$, one can instead minimise $-f(x)$. The function to be optimised is typically called the objective function. However, in the context of minimisation, it may also be referred to as the cost function, loss function, or error function.

To determine how to adjust x to optimise $f(x)$, one requires the derivative of the function, denoted $f'(x)$ or $\frac{df}{dx}$, which gives the slope of $f(x)$ at the point x , as explained by Goodfellow et al. [57]. This slope indicates how small changes in the input x affect the output $f(x)$ and can be used to guide adjustments as shown in eq. (2.8).

$$f(x + \epsilon) \approx f(x) + \epsilon f'(x) \quad (2.8)$$

This method, known as gradient descent, was introduced in 1847 by Cauchy [71], as cited by Goodfellow et al. [57], and it is illustrated in fig. 2.6. Gradient descent uses the derivative to adjust x in the direction that reduces $f(x)$.

However, when $f'(x) = 0$, the derivative provides no information about the direction of change, as pointed out by Goodfellow et al. [57]. These points are called critical points or stationary points, and they can be local minima, local maxima, or saddle points. A global minimum occurs when $f(x)$ reaches its absolute lowest value.

According to Goodfellow et al. [57], in deep learning, the functions to be optimised often have numerous local minima and saddle points, which make the optimisation process more

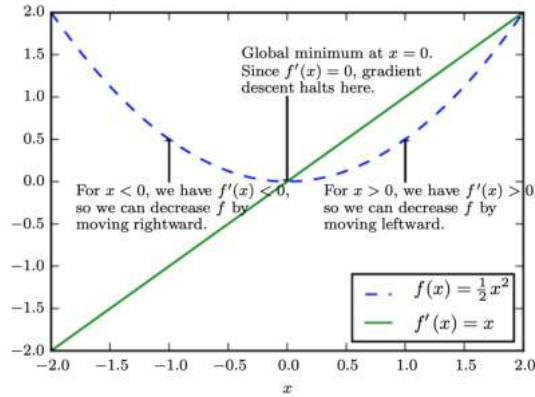


Figure 2.6: An illustration of gradient descent by Goodfellow et al. [57].

challenging. These functions are typically high-dimensional, so finding the global minimum is rarely feasible. Instead, practical optimisation often involves finding a sufficiently low value of $f(\mathbf{x})$, even if it is not the true minimum.

When dealing with functions with multiple inputs, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, it is essential to consider the concept of partial derivatives, as Goodfellow et al. [57] point out. The partial derivative $\frac{\partial}{\partial x_i} f(\mathbf{x})$ quantifies how the function f changes when only the variable x_i increases at the point $\mathbf{x}$. The gradient extends the idea of the derivative to cases involving vectors: the gradient of f is the vector containing all the partial derivatives, denoted $\nabla_{\mathbf{x}} f(\mathbf{x})$. The i -th element of the gradient represents the partial derivative of f with respect to x_i .

Goodfellow et al. [57] further explain that critical points in multiple dimensions are identified where every gradient element equals zero. To explore the behaviour of the function at these points, one can define the directional derivative in the direction of a unit vector $\mathbf{u}$ as the slope of the function f in that direction. To minimise f , one must determine the direction in which f decreases the fastest, achievable by aligning $\mathbf{u}$ with the negative gradient. Thus, the gradient points uphill while the negative gradient points downhill, indicating that f decreases by moving towards the negative gradient.

The gradient descent algorithm suggests a new point as given in eq. (2.9), according to Goodfellow et al. [57], where ϵ represents the learning rate, a positive scalar determining the step size. Convergence in gradient descent occurs when every gradient element is zero or close to zero.

$$\mathbf{x}' = \mathbf{x} - \epsilon \nabla_{\mathbf{x}} f(\mathbf{x}) \quad (2.9)$$

Goodfellow et al. [57] highlight that most deep learning relies on a crucial extension of gradient descent known as Stochastic Gradient Descent (SGD). The key insight of SGD is that the gradient can be treated as an expectation, which can be approximated using a small set of samples. In each step of the algorithm, a mini-batch of examples $\mathcal{B} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m')}\}$ is uniformly sampled from the training set. The gradient estimate is calculated as show in eq. (2.10), where m' is the mini-batch size and $\mathcal{L}(\mathbf{x}^{(i)}, y^{(i)}, \boldsymbol{\theta})$ is the per-example loss function parameterised by $\boldsymbol{\theta}$.

$$\mathbf{g} = \frac{1}{m'} \nabla_{\boldsymbol{\theta}} \sum_{i=1}^{m'} \mathcal{L}(\mathbf{x}^{(i)}, y^{(i)}, \boldsymbol{\theta}), \quad (2.10)$$

Subsequently, the SGD algorithm follows the estimated gradient downhill and adjusts the parameters as $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \epsilon \mathbf{g}$.

In the context of gradient descent, momentum is another essential concept to consider; however, a comprehensive explanation of this concept falls outside the scope of our current

work. For further insights into momentum, we recommend referring to Chapter 8.3.2 in the work by Goodfellow et al. [57].

The chain rule in calculus is a fundamental concept used to compute the derivatives of composite functions, which is crucial for training neural networks, as pointed out by Goodfellow et al. [57]. Backpropagation [62], the algorithm employed for this task, computes the chain rule efficiently by following a specific order of operations.

Consider $x \in \mathbb{R}$, with $f, g : \mathbb{R} \rightarrow \mathbb{R}$. Suppose $y = g(x)$ and $z = f(g(x)) = f(y)$, representing a two-layer network. The chain rule states that

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}. \quad (2.11)$$

This can be generalised beyond the scalar case. If $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$, then the chain rule can be expressed as

$$\frac{\partial z}{\partial x_i} = \sum_j \frac{\partial z}{\partial y_j} \frac{\partial y_j}{\partial x_i}. \quad (2.12)$$

In vector notation, this can be equivalently written as

$$\nabla_{\mathbf{x}} z = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^T \nabla_{\mathbf{y}} z, \quad (2.13)$$

where $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$ is the $n \times m$ Jacobian matrix containing all the partial derivatives of g . This formulation shows that the gradient of a variable $\mathbf{x}$ can be obtained by multiplying the Jacobian matrix by the gradient $\nabla_{\mathbf{y}} z$. For a thorough description and examples of these concepts, refer to Goodfellow et al. [57] and Russell & Norvig [64], who provide in-depth explanations and applications related to neural networks.

2.7 Image Restoration

In digital image processing, image restoration encompasses various techniques to improve data's visual quality and clarity. As digital images often suffer from degradation due to noise, blur, and resolution limitations, restoration methods are crucial for enhancing the visual fidelity of these images.

2.7.1 Image Quality Assessment

According to Moser et al. [72], numerous factors contribute to exceptional image quality, including sharpness, contrast, and noise minimisation. As a result, accurately evaluating image restoration models poses a significant challenge. Here, we present our evaluation methods, categorised under the term IQA. IQA encompasses any metric that relies on the perceptual evaluations of human viewers, gauging how authentic the image appears after restoration, as explained by Moser et al. [72]. IQA can include subjective assessments (e.g., human raters) or objective metrics (e.g., formal evaluations). Lepcha et al. [69] add that while subjective assessments are more aligned with perceptual requirements, they can be time-consuming and expensive, making objective metrics the preferred standard. However, these techniques are not always consistent with each other, as objective methods often struggle to accurately capture human visual perception, leading to significant discrepancies in IQA results.

The most simple measurement is the *Mean Absolute Error* (MAE), which measures the average absolute difference between the predicted and ground truth pixel values:

$$\text{MAE}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N_{\mathbf{y}}} \sum_{p \in \Omega_{\mathbf{y}}} | \mathbf{y}_p - \hat{\mathbf{y}}_p | \quad (2.14)$$

The Poisson loss is frequently employed for data following a Poisson distribution, especially in scenarios where the variance is dependent on the mean, such as photon counts in astronomical imaging, according to Sweere et al. [1]. This type of loss measures the likelihood of the generated pixel values, given that the target originates from a Poisson distribution conditioned on the input. The Poisson loss is defined as:

$$\text{Poisson}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N_{\mathbf{y}}} \sum_{p \in \Omega_{\mathbf{y}}} \hat{\mathbf{y}}_p - \mathbf{y}_p \log(\hat{\mathbf{y}}_p) \quad (2.15)$$

One popular objective quality measurement is the *Peak Signal-to-Noise Ratio* (PSNR). It is the ratio between the maximum possible pixel value p_{max} (255 for 8-bit representations) and the *Mean Squared Error* (MSE) of reference images. Given the approximation $\hat{\mathbf{y}}$ and the ground truth $\mathbf{y}$, PSNR is a logarithmic quantity using the decibel scale:

$$\text{PSNR}(\mathbf{y}, \hat{\mathbf{y}}) = 10 \log_{10} \left(\frac{p_{max}^2}{\text{MSE}(\mathbf{y}, \hat{\mathbf{y}})} \right), \text{ with} \quad (2.16)$$

$$\text{MSE}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N_{\mathbf{y}}} \sum_{p \in \Omega_{\mathbf{y}}} (\mathbf{y}_p - \hat{\mathbf{y}}_p)^2 \quad (2.17)$$

Even though PSNR is commonly used as an evaluation measure, it often produces unsatisfactory results in real-world situations because it emphasises pixel-level differences rather than considering human visual perception, which is more sensitive to image structures, according to Moser et al. [72]. As a result, new metrics have been developed to focus on structural features within an image like *Structural Similarity Index Measure* (SSIM) [73]. The SSIM metric considers three relatively independent components, luminance, contrast, and structure, and is defined as

$$\text{SSIM}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{C}_l(\mathbf{y}, \hat{\mathbf{y}})^\alpha \cdot \mathcal{C}_c(\mathbf{y}, \hat{\mathbf{y}})^\beta \cdot \mathcal{C}_s(\mathbf{y}, \hat{\mathbf{y}})^\gamma. \quad (2.18)$$

Here, $\mathcal{C}_l$ is the luminance and $\mathcal{C}_c$ the contrast comparison based on the similarity comparison function S introduced by Wang et al. [73], $\mathcal{C}_s$ is the structure comparison, given by the correlation coefficient between $\mathbf{y}$ and $\hat{\mathbf{y}}$, and $\alpha, \beta, \gamma > 0$ are adjustable control parameters for weighting the relative importance of all components, as explained by Moser et al. [72].

According to Moser et al. [72], it is common to enhance images to different scaling factors when using SR models through a technique known as Multi-Scaling (MS). As a result, when evaluating image quality, it is important to consider this scenario. The *Multi-Scaling-SSIM* (MS-SSIM) [74] is an extension of the SSIM metric that accounts for MS. It is more resilient to changes in viewing conditions than SSIM and includes parameters to weigh the importance of different scales. To calculate MS-SSIM, a low-pass filter is applied, and the filtered image is downsampled by a factor of 2 for each scale level from 1 to the largest scale, denoted as s_{max} . This allows for the calculation of contrast and structure comparisons at each scale and the luminance comparison for the largest scale. The MS extension changes SSIM to

$$\text{MS-SSIM}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{C}_l(\mathbf{y}, \hat{\mathbf{y}})^\alpha \cdot \prod_{i=1}^{s_{max}} \mathcal{C}_{c,i}(\mathbf{y}, \hat{\mathbf{y}})^{\beta_i} \cdot \mathcal{C}_{s,i}(\mathbf{y}, \hat{\mathbf{y}})^{\gamma_i}. \quad (2.19)$$

Moser et al. [72] point out that some metrics utilise predefined features, such as the *Feature Similarity Index* (FSIM) [75], which combines gradient magnitudes with Phase Congruency (PC), a local structure measurement, as feature points. The PC model posits a biologically plausible explanation for how the human visual system detects and identifies image features. These features are perceived where the Fourier components are maximal in phase. Zhang et al. [75] define FSIM as

$$\text{FSIM}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\sum_{p \in \Omega_{\mathbf{y}}} S_L(p) \cdot \text{PC}_m(p)}{\sum_{p \in \Omega_{\mathbf{y}}} \text{PC}_m(p)}, \quad (2.20)$$

where S_L is the similarity measure between the two pictures based on their similarity in PC and gradient magnitude features, and $PC_m = \max(PC(\mathbf{y}), PC(\hat{\mathbf{y}}))$.

An extension to FSIM is the *Haar wavelet-based Perceptual Similarity Index* (HaarPSI) [76], which employs a Haar wavelet decomposition [77] to evaluate local similarities. According to its creators, HaarPSI outperforms state-of-the-art IQA methods such as SSIM and FSIM in terms of both execution time and higher correlations with human opinion scores. They propose that a multi-scale complex-valued wavelet filter bank directly impacts the computation of FSIM when calculating the Fourier components. It is defined as

$$\text{HaarPSI}(\mathbf{y}, \hat{\mathbf{y}}) = \sigma_\alpha^{-1} \left(\frac{\sum_{p \in \Omega_{\mathbf{y}}} \sum_{k=1}^2 \text{HS}(\mathbf{y}, \hat{\mathbf{y}}) \cdot \mathbf{W}(\mathbf{y}, \hat{\mathbf{y}})}{\sum_{p \in \Omega_{\mathbf{y}}} \mathbf{W}(\mathbf{y}, \hat{\mathbf{y}})} \right)^2. \quad (2.21)$$

Here, $\sigma_\alpha^{-1}(x)$ is the inverse sigmoid function, where x is weighted by a constant $\alpha > 0$, $\text{HS}(\mathbf{y}, \hat{\mathbf{y}})$ is the local similarity measure based on the Haar wavelet transform and $\mathbf{W}(\mathbf{y}, \hat{\mathbf{y}})$ is a weight map derived from the response of a single low-frequency Haar wavelet filter.

2.7.2 Super-Resolution

The resolution of an image significantly impacts its detail and usability, particularly in fields such as medical imaging, satellite imagery, and consumer photography. Super-Resolution (SR), as described by Anwar et al. [78] and Moser et al. [72] involves enhancing a given Low-Resolution (LR) image $\mathbf{x}$ with limited details to produce a High-Resolution (HR) image $\mathbf{y}$

$$\mathbf{x} \in \mathbb{R}^{w_{\mathbf{x}} \times h_{\mathbf{x}} \times c}, \mathbf{y} \in \mathbb{R}^{w_{\mathbf{y}} \times h_{\mathbf{y}} \times c}, w_{\mathbf{x}} < w_{\mathbf{y}}, h_{\mathbf{x}} < h_{\mathbf{y}}, \quad (2.22)$$

resulting in improved visual quality and refined details. The process can involve using a single image (Single Image Super-Resolution (SISR)) or multiple images (Multiple Image Super-Resolution (MISR)).

We primarily focus on SISR, which remains a complex and open research problem in computer vision for several reasons, as highlighted by Anwar et al. [78]. Firstly, it presents an ill-posed inverse problem, meaning multiple solutions can exist for the same low-resolution image instead of a single unique solution. To address this, reliable prior information is typically necessary to constrain the solution space. Secondly, as the upscaling factor increases, the problem becomes more complex. At higher factors, recovering missing scene details becomes increasingly challenging, potentially leading to the introduction of incorrect information. Lastly, assessing the quality of the output is not straightforward; for instance, quantitative metrics such as PSNR and SSIM only loosely correlate with human perception.

Anwar et al. [78] note that SR methods generally fall into two main categories: traditional and deep learning. While traditional algorithms have been in use for many years, they have been surpassed in performance by deep learning-based methods. Consequently, modern algorithms predominantly utilise deep learning models driven by data to accurately reconstruct the necessary details for SR.

Throughout this work, we use the same notation as Moser et al. [72]. Therefore, $N_{\mathbf{x}}$ denotes the number of pixels in an image $\mathbf{x}$ and $\Omega_{\mathbf{x}}$ represents the set of all valid positions in $\mathbf{x}$:

$$N_{\mathbf{x}} = w_{\mathbf{x}} \cdot h_{\mathbf{x}} \cdot c \quad (2.23)$$

$$\Omega_{\mathbf{x}} = \{(i_{\mathbf{x}}, j_{\mathbf{x}}, k_{\mathbf{x}}) \in \mathbb{N}^3 \mid i \leq w_{\mathbf{x}}, j \leq h_{\mathbf{x}}, k_{\mathbf{x}} \leq c\} \quad (2.24)$$

Let $s \in \mathbb{N}$ be a scaling factor and $\mathcal{D}$ be a degradation mapping that describes the inherent relationship between $\mathbf{x}$ and $\mathbf{y}$:

$$h_{\mathbf{y}} = s \cdot h_{\mathbf{x}} \text{ and } w_{\mathbf{y}} = s \cdot w_{\mathbf{x}} \quad (2.25)$$

$$\mathcal{D} : \mathbb{R}^{w_{\mathbf{y}} \times h_{\mathbf{y}} \times c} \rightarrow \mathbb{R}^{w_{\mathbf{x}} \times h_{\mathbf{x}} \times c} \quad (2.26)$$

$$\mathbf{x} = \mathcal{D}(\mathbf{y}, \delta), \quad (2.27)$$

where δ represents the parameters of $\mathcal{D}$ containing, for example, the scaling factor s and other elements like blur type. In practice, the degradation mapping is often unknown and therefore approximated, for instance, with bicubic downsampling. The primary challenge of SISR is to perform the inverse mapping of $\mathcal{D}$. The objective is to find a SR model $\mathcal{M}_{SR}$:

$$\mathcal{M}_{SR} : \mathbb{R}^{w_{\mathbf{x}} \times h_{\mathbf{x}} \times c} \rightarrow \mathbb{R}^{w_{\mathbf{y}} \times h_{\mathbf{y}} \times c}, \text{ so that} \quad (2.28)$$

$$\hat{\mathbf{y}} = \mathcal{M}_{SR}(\mathbf{x}, \theta), \quad (2.29)$$

where $\hat{\mathbf{y}}$ is the predicted HR approximation of the LR image $\mathbf{x}$ and θ are the parameters of $\mathcal{M}$. In the context of deep learning, this translates to an optimisation objective that minimises the difference between the estimation $\hat{\mathbf{y}}$ and the ground truth HR image $\mathbf{y}$ using a given loss function $\mathcal{L}$:

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(\mathbf{y}, \hat{\mathbf{y}}, \theta) \quad (2.30)$$

2.7.3 Denoising

Noise is an inherent problem in digital imaging, often arising from environmental factors or the limitations of the imaging sensors. DN is the process of removing noise from an image to recover the original scene as faithfully as possible, as defined by Izadi et al. [79]. Considering a noisy image $\mathbf{x} \in \mathbb{R}^{w \times h \times c}$ with $N_{\mathbf{x}}$ pixels corrupted by a degradation function $\mathcal{D}$, and its corresponding clean counterpart $\mathbf{y} \in \mathbb{R}^{w \times h \times c}$ with $N_{\mathbf{y}} = N_{\mathbf{x}}$ pixels. The degradation function $\mathcal{D}$ can be expressed as:

$$\mathbf{x} = \mathcal{D}(\mathbf{y}, \delta), \quad (2.31)$$

where δ represents the degradation functions' parameters and noise model. Noise corruption is often modelled as noise addition, followed by pixel-wise clipping to account for sensor saturation. Let $\boldsymbol{\eta} \in \mathbb{R}^{w \times h \times c}$ represent the noise component for every pixel in $\mathbf{y}$. The additive noise model can be written as

$$\mathbf{x} = \text{clip}(\mathbf{y} + \boldsymbol{\eta}), \text{ with} \quad (2.32)$$

$$\text{clip}(\mathbf{z}) = \min(\max(\mathbf{z}, 0), 1), \quad (2.33)$$

assuming the pixel intensities are in the range 0 to 1. DN aims to recover $\mathbf{y}$ from the observed noisy data $\mathbf{x}$. Typically, the degradation function and the noise parameters are unknown, according to Izadi et al. [79]. The objective is to find a DN model $\mathcal{M}_{DN}$:

$$\mathcal{M}_{DN} : \mathbb{R}^{w \times h \times c} \rightarrow \mathbb{R}^{w \times h \times c}, \text{ such that} \quad (2.34)$$

$$\hat{\mathbf{y}} = \mathcal{M}_{DN}(\mathbf{x}, \theta), \quad (2.35)$$

where $\hat{\mathbf{y}}$ is the predicted noise-less approximation of the noisy image $\mathbf{x}$ and θ are the parameters of $\mathcal{M}_{DN}$. Similar to SR, this corresponds to an optimisation goal that minimises the difference between the estimation $\hat{\mathbf{y}}$ and the ground truth image $\mathbf{y}$ using a given loss function $\mathcal{L}$:

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(\mathbf{y}, \hat{\mathbf{y}}, \theta) \quad (2.36)$$

Izadi et al. [79] add that like SR, DN is an ill-posed inverse problem and poses the same challenging assessment of the output's quality.

3 Related Work

The following chapter provides an overview of the current progress in SR and DN models applied to two distinct types of images. The rationale for this division is straightforward: most advancements in these tasks have concentrated on optical images. Consequently, section 3.1 examines the key developments and techniques that have shaped the field for these widely studied image types. However, given that this thesis is concerned with enhancing images captured by the XMM-Newton X-ray satellite, section 3.2 shifts the focus to related work in the specialised domain of X-ray imaging.

3.1 Enhancing Optical Images

According to Moser et al. [72], CNN-based architectures are straightforward to comprehend and typically require minimal computational resources due to their small size. Many of these architectures were prominent in the early stages of deep learning-based SR since their performance lags behind the state-of-the-art. Additionally, the „the deeper, the better“ principle of deep learning does not align well with simple networks due to the issue of vanishing/exploding gradients [80]. The following paragraph summarises the findings of Moser et al. [72]. The initial CNN developed for SR was SRCNN, introduced by Dong et al. [81]. In a subsequent paper, the authors explored methods to accelerate SRCNN, leading to the development of FSRCNN [82], which incorporated key optimisations to improve performance and significantly reduce computation time. Shi et al. [83] proposed a model called ESPCN, where the network’s last layer performs the upscaling. Therefore, each LR image is directly input into the network, and feature extraction occurs through nonlinear convolutions in LR space. The reduced input resolution allows the use of a smaller filter size, effectively integrating the same information while maintaining a given contextual area. This reduction in resolution and filter size significantly lowers the computational and memory complexity, enabling real-time super-resolution of videos. [83] The SRGAN, proposed by Ledig et al. [84], is considered the benchmark architecture for GAN-based SISR. The method employs an adversarial objective feature to encourage SR outputs that closely resemble natural images. This architecture favours the generation of outputs that are perceptually similar to HR images. Building on SRGAN, Wang et al. [85] introduced ESRGAN, which features a deeper and more complex structure. ESRGAN uses the Residual-in-Residual Dense Block (RRDB) as the basic building block with dense connections and multi-level residual networks, resulting in improved texture recovery and an enhanced capacity to capture semantic information while reducing unwanted artefacts in

the output images. Furthermore, ESRGAN addresses issues such as mode collapse and vanishing gradients, which were present in SRGAN. For further details about mode collapse and other challenging issues with GANs, Singla et al. [86] provide comprehensive insights.

Transformer-based architectures continue to demonstrate remarkable reconstruction performance, according to Chen et al. [87], as evidenced by models such as SwinIR [88], HAT [89], and DAT [90]. Since transformers were initially designed for Natural Language Processing (NLP), they are naturally suited to work with data sequences. When applied to images, this means dividing the image into patches, a notable limitation for using transformers in SR tasks. SwinIR addresses this issue by alternating between attention blocks to facilitate the transfer of information across patches. This architecture adheres closely to the transformer concept, specifically through multiple Swin transformer [91] layers incorporating multi-head self-attention [68]. SwinFIR [92] builds on this concept by incorporating fast Fourier convolutions. HAT [89] architecture merges channel attention and self-attention mechanisms to leverage the strengths of both: channel attention’s ability to utilize global information and self-attention’s strong representational power. Additionally, they incorporate an overlapping cross-attention module to facilitate more direct interaction between features in adjacent windows. These design choices allow the model to engage more pixels during reconstruction, leading to a notable enhancement in performance. DAT [90] enhances representation capability by dually aggregating spatial and channel features between and within blocks. Through extensive experiments, they show that DAT surpasses state-of-the-art methods while maintaining lower complexity and smaller model size.

The field of super-resolution is rapidly evolving, and various model types are being developed. Some examples of these models include flow-based models, diffusion models, neighbour embedding, and sparse coding. For a broader overview, we refer to the reviews by Moser et al. [72,93] and Lepcha et al. [69].

Using neural networks for DN gained traction in 2008 when Jain and Seung [94] proposed leveraging convolutional layers to reduce the computational expenses associated with parameter estimation and inference in popular probabilistic DN methods. Zhang et al. [95] were among the first to address spatially varying noise by adapting the performance of a plain deep CNN for different regions. They achieved this by augmenting the noisy image with a noise level map before feeding it into the network. These noise level maps were generated by stretching the actual or estimated noise variance across spatial dimensions to match the input size. In contrast to using clean images, some DN methods utilise long skip connections in their design to learn the residue between the noisy and clean images. Notably, DnCNN by Zhang et al. [96] was one of the earliest attempts in this direction and showed superior performance with simpler architectures. The residue image produced in the output of the DnCNN is then subtracted from the input noisy image to obtain the clean estimate. Instead of learning a complex mapping from one complete image to another, DnCNN learns the residue image to remove the noise and recover the high-frequency details. Xie et al. [97] introduced a Stacked Sparse DN Autoencoder (SSDA) framework, which involved sequentially stacking multiple instances of DN autoencoders to connect the noisy input to the network output. Building upon this work, Agostinelli et al. [98] extended the SSDA framework by introducing adaptive multi-column SSDA to enhance its robustness against various types of noise. Additionally, Mao et al. [99] proposed an early exploration of using residual connections for image DN in REDNet, where skip connections were employed between corresponding layers in a mirrored autoencoder architecture. Anwar & Barnes [100] introduced the first work to take advantage of the growing popularity of the channel-wise attention mechanism. In a similar vein, Zamir et al. [101,102] suggested integrating a supervised attention module between each pair of stages in a multi-stage architecture. Moreover, they introduced a cross-stage information exchange module to en-

hance feature fusion between early and later stages. Chen et al. [103] was the first to utilise GANs for real noise modelling and to create a paired training dataset. In their approach, the generator produces synthetic noise, while the discriminator is trained to differentiate between real and synthetic noise. However, this method only uses a random vector as the input to the noise generator, resulting in noise samples that are independent of the signal, as the network has not been exposed to the clean intensity signal during training. More recently, Chang et al. [104] proposed a method to separate the noise generation process from the camera characteristics using two distinct networks. Specifically, a noise generator processes the clean image and an initial noise sample. Additionally, a latent vector generated from a camera encoding network is used to transfer camera-specific characteristics of the image to the noise generator through feature concatenation. The final synthetic noise is obtained from the output of the generative network.

A more comprehensive overview of available deep learning-based DN methods can be found in Gu & Timofte [105], Tian et al. [106], and Izadi et al. [79]. An overview of techniques not based on deep learning can be found in Goyal et al. [107].

While deep learning techniques have significantly addressed SR and DN as separate tasks, the joint problem of SR and DN has yet to be as extensively explored. The joint task is particularly challenging because it requires balancing the trade-off between enhancing image resolution and effectively removing noise without introducing artefacts or losing important details. After separately training the DN and SR networks, Bei et al. [108] combine them and continue training them as a single network in two different ways. DNISR first runs the image through DnCNN [96] to produce a LR noise-reduced image and then passes the result through EDSR [109], a SR network, to generate a HR image. On the other hand, DNSR uses a more complex concatenation procedure. It eliminates the information bottleneck between the two networks by combining the DN tail layer and the SR network's head layer into a single bridge convolutional layer. This layer directly maps the number of feature maps in the DN to the number of feature maps in the SR network. In their paper, Zamir et al. [101] introduce MIRNet, an architecture that utilises their novel Multi-Scale Residual Block (MRB). Initially, the network employs a convolutional layer to capture low-level features. Subsequently, multiple MRBs are used to extract deep features. Finally, a convolutional layer is applied to extract a residual image, which is then added to the input image to produce the restored image. In their work, Villar-Corrales et al. [110] and Cammarasana & Patanè [111] both utilize WDSR [112] with distinct objectives. The former delves into the comparison between deploying DN and SR networks sequentially and integrating both at the feature level. Their experiments reveal that each variant offers specific advantages: the feature-level design achieves the best results when the type of image corruption aligns in both the training and testing datasets, regardless of the choice of denoiser. On the other hand, the sequential design demonstrates superior performance on unseen types of image corruption, addressing a critical drawback of existing super-resolution models. In contrast, Cammarasana & Patanè [111] employ WDSR to generate HR images without altering the noise distribution. Applying filters to noisy images without altering the noise component is crucial for preserving information, maintaining input image quality, and improving the robustness of post-processing algorithms.

The existing literature on the joint task typically extends models initially designed for either SR or DN, adapting them to perform both tasks simultaneously. Despite these efforts, the joint task still needs to be explored, with much room for innovation and improvement.

3.2 Enhancing Astronomical Images

The most present image restoration approaches for astronomical imagery are based on the GAN [67] architecture. Schawinski et al. [113] chose to use the standard GAN architecture

to denoise galaxy images well beyond the capabilities of a simple deconvolution. Castro [114] used the approach of Real-ESRGAN [115], which is an extension of ESRGAN [85] and utilizes the same generator, which is a deep network with RRDBs. The upsample layers employ nearest-neighbor interpolation, an U-Net [116] with spectral normalization [117] is used as the discriminator. Sweere et al.'s [1] SR model is based solely on the RRDB generator from ESRGAN. They have also replaced the nearest-neighbour interpolation up-sampling layer with pixel shuffle upsampling, following the approach by Shi et al. [83]. Pixel shuffle offers more connections and does not interpolate the upsampled image, which should enhance the quantitative details on smaller scales. This modified architecture produces more dependable outputs but can only create precise reconstructions in areas where the model has learned to generate highly confident features, such as high-signal-to-noise point sources. When it comes to aspects with low confidence, like background noise, this will lead to blurry reconstructions, and the output image is unlikely to resemble the target closely. Additionally, the areas of low and high confidence vary for each image since they depend on the features present in the input image. Their DN model has a similar architecture to their SR model, however without the upsampling layer and the last convolutional layer. They also incorporated more skip connections, drawing inspiration from Zhang et al. [118]. This modification proved to be beneficial in learning smaller features within the image, ultimately enhancing training speed. The model no longer needs to process all the small features, but rather learns the features to suppress in order to generate the de-noised output image. Lauritsen et al. [119] use an autoencoder-based architecture based on the generator network of GALAXYGAN [113] and a flux based loss function. The data consists of simulated and real observations. The network processes both sets in training independently and in succession. Each epoch starts by training on the complete simulated training set, followed by three training rounds on the observed data set. The goal was to have the network learn the key structural features in the simulated data before fine-tuning it using the observed data to handle any slight differences between observed and simulated images. Furthermore, their goal was to super-resolve the PSF instead of increasing the target image size or decreasing the pixel size. A finer PSF allows to differentiate objects more easily.

The overwhelming presence of GAN-based architectures, does not mean that no other approaches were tried. In their research, Vojtekova et al. [120] focus on U-net [116]. Their default architecture is based on the work of Chen et al. [121], with adjustments made to the number of input and output channels to fit their specific data. The U-Net offers the advantage of accommodating input images of varying sizes, although processing large images may demand a substantial amount of GPU memory. During the evaluation phase, they maintain the same input size used during training. To reconstruct the full image size, they employ a mosaic approach, extracting image crops of dimensions 256×256 , passing them through the network, shifting by 32 pixels, and repeating the process. In areas of overlap, they compute the mean pixel value. Soltis et al. [122] have implemented an architecture utilizing CNN to remove noise from X-ray images captured by the eROSITA satellite. The architecture comprises three paths, each corresponding to different energy bands and the unique properties of astrophysical objects captured in these bands. In their paper, Park et al. [123] propose using Zamir et al.'s [3] efficient transformer for DN to improve astronomical images. Specifically, they are exploring the possibility of enhancing Hubble Space Telescope images to achieve the same quality as James Webb Space Telescope images regarding both resolution and depth. They initially pre-trained the model using simplified galaxy images and then fine-tuned it using realistic galaxy images.

For more insight on different applications of Deep Learning (DL) on astronomical data, please refer to Smith & Geach [124].

4 Datasets

Given our supervised approach for both SR and DN, we require paired datasets: LR (with or without noise) and HR (without noise) images for SR, and high-noise/low-noise pairs for DN. Extracting such pairs from actual XMM-Newton images is impossible since noise is always present (see section 2.4.3), and XMM’s sensors cannot capture higher resolution images. According to Sweere et al. [1], an *SR dataset* is typically created by digitally downsampling HR images or aligning images taken with different optical zoom scales. However, our goal is to generate images with higher spatial resolution, featuring smaller pixel sizes and PSF, as also aimed by Sweere et al. [1]. Thus, downsampling is not viable as it only decreases pixel size without affecting the PSF, and zooming is unfeasible since the XMM-Newton telescope lacks that capability. While integrating data from other X-ray telescopes like Chandra might help, the limited overlap in observed fields prevents effective training for an SR model, as stated by Sweere et al. [1].

In contrast, an *DN dataset* is often created synthetically by adding Gaussian noise, such as with Set12 [96], BSD68 [125], and Urban100 [126], or by estimating low-noise images using multiple exposures (as in SIDD [127]) or adjusting ISO settings (as in DND [128]). However, none of these approaches are suitable since the noise captured by XMM-Newton does not resemble Gaussian noise. The noise is highly variable, depending on the spacecraft’s position and other astronomical factors, and there is no way to effectively isolate or remove it from the observations, as described in section 2.4.3.

Therefore, we employ the same simulation framework as Sweere et al. [1] to artificially enhance the resolution and generate a training dataset that preserves the essential observational properties of actual XMM-Newton images. Our simulation process builds on the work of Sweere et al. [1], with additional features that allow individual simulation of each CCD and various minor usability and performance improvements. The real observations gathered by Sweere et al. [1] from XMM-Newton are used exclusively for visual evaluation and not for model training.

In Section 4.1, we briefly introduce the dataset of real observations. Section 4.2 describes the simulation procedure. Section 4.3 outlines our pre-processing steps, and Section 4.4 details how we divide the simulated data into training, validation, and test sets.

4.1 Real XMM-Newton Observations

According to the XMM-Newton User Handbook (UHB), every observation conducted by XMM-Newton is accessible through the XMM-Newton Science Archive (XSA). This

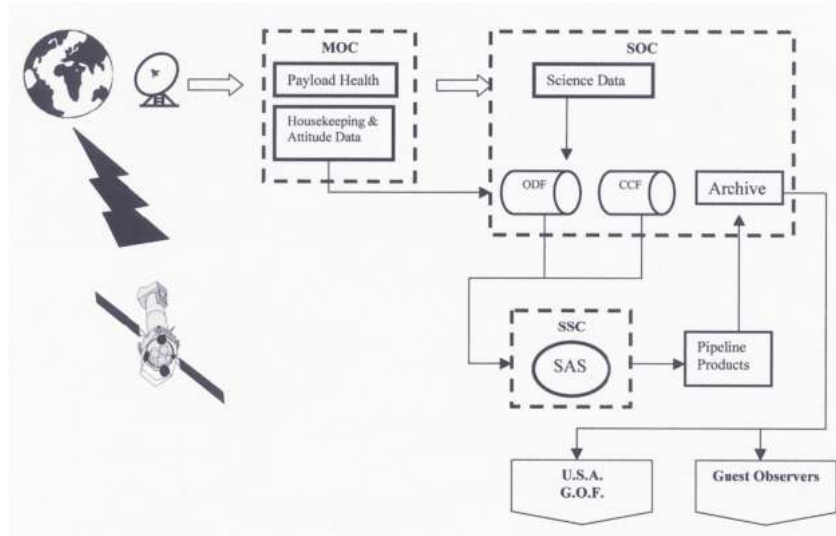


Figure 4.1: The data flow for every observation by XMM-Newton, as illustrated by Jansen et al. [39].

archive contains Observation Data Files (ODFs), generated from spacecraft information data and each active instrument’s raw, uncalibrated science data. These ODFs are subsequently processed by the Processing Pipeline Subsystem (PPS), which automatically generates PPS products for each instrument mode by running specific tasks within the Science Analysis System (SAS). One of these products is an event list containing information about photon impacts, similar to the event file created by SIXTE. A set of calibration files, known as Current Calibration Files (CCFs), is essential for this process. These calibration files contain all relevant details about the telescopes, instruments, and the spacecraft. Figure 4.1 provides a superficial overview of the data flow involved in this process.

The observations were made using the EPIC-pn instrument, utilising all twelve of its CCDs with the thin filter. The energy band for the dataset ranges from 0.5 keV to 2.0 keV, and the entire XSA was considered at the time of creation. Observations shorter than 20 ks, or those containing bad time intervals and events, were filtered out. Event lists from the observations were split into multiples of 10 ks intervals. For example, a 50 ks exposure was divided into five 10 ks segments, two 20 ks segments, one 40 ks, and one 50 ks segment.

Filters were applied to reduce contamination from background events and bad or dead pixels. Only photon events with well-calibrated patterns were retained, and although out-of-FOV events were initially kept to study the background, we removed these events since our focus excludes them. The cleaned event lists generated images in detector coordinates, which were saved in FITS format.

The final dataset consists of 5554 unique exposures, producing 23 799 sub-images. Figure 4.2 contains some example images. We use this dataset solely to visualise the performance of our models on real data and do not employ it for training purposes.

4.2 Simulated Dataset

We generate two sets of simulated images: one using the actual PSF of the instruments, and the other using a PSF rescaled to twice the resolution, i.e., the FWHM and HEW are halved. Additionally, the pixel size is reduced by half, which keeps the PSF sampling the same as the original but results in four times more pixels for the same FOV. The detailed process of defining the detectors is outlined in section 4.2.1.

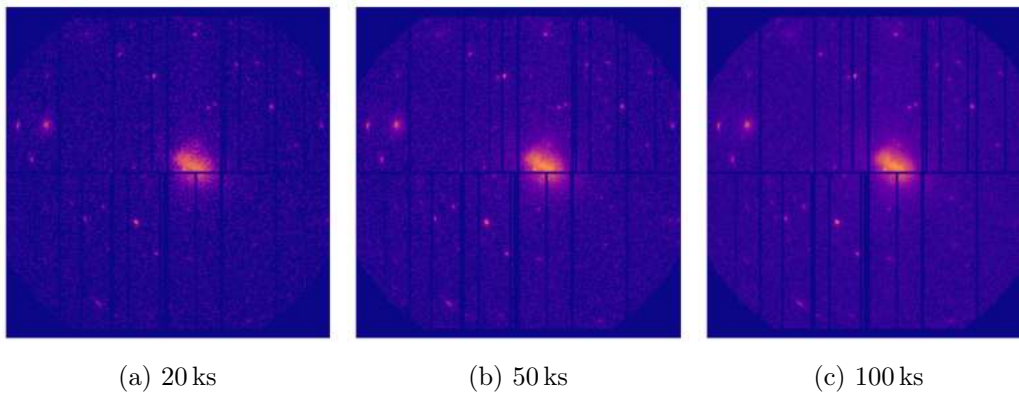


Figure 4.2: Example images from the real dataset with different exposure times for the observation 0503490201. Images use the square root scaling and `matplotlib`'s [129] `plasma` colourmap.

Our focus lies in simulating observations of complex astronomical fields, such as galaxies or galaxy clusters. These images comprise three primary components: the extended emission from the galaxy or cluster, contaminant point sources (primarily AGN), and the background. Each of these components, along with their simulation methods, is described in section 4.2.2.

The repository containing the code for generating the necessary files for SIXTE and running the simulations can be found at <https://github.com/SamSweere/xmm-epicpn-simulator>. Although our simulation process is based on the work of Sweere et al. [1], we have incorporated more realistic properties, such as the entire energy spectrum of XMM-Newton and automated simulations without the need for manual removal of faulty results.

4.2.1 Describing the Detectors in SIXTE

As outlined in section 2.5, to adequately describe the EPIC-pn, EPIC-MOS1, and EPIC-MOS2 instruments, several files are required: RMF, ARF, files detailing the PSF and vignetting, and XMLs that contain descriptions of the detectors. Except for the XMLs, these files are provided by the SIXTE team and are based on the CCFs. The PSF and vignetting files were calculated by Sweere et al. [1], and we use them as given, except for the PSF file. When simulating at double the resolution, we modify the PSF file by halving the pixel size to match the increased resolution.

The code for generating the final three files mentioned automatically uses the most recent CCFs available locally, meaning that the simulation results may vary depending on the version of the CCFs used. For more detailed information on the available CCFs, please refer to the work of Altieri et al. [130].

The procedure for creating the XML files for the instruments follows the same steps for each. First, we obtain the necessary properties of the telescopes and detectors, such as the focal length, pixel sizes (in metres), and the FOV (in degrees). Unlike Sweere et al. [1], we use the correct FOV rather than a slightly larger one. Additionally, we define the properties of the respective CCDs, including the time required to read out each pixel and their orientation. Figure 4.3 shows how SIXTE sees each instrument.

As noted in section 2.4, the detectors consist of multiple CCDs. For the simulation, one can opt to simulate each CCD individually or simulate a single, larger CCD, applying a mask to replicate the gaps and inactive areas of the detector. The preferred approach is to individually simulate each CCD to maintain fidelity with the real instruments. Furthermore,

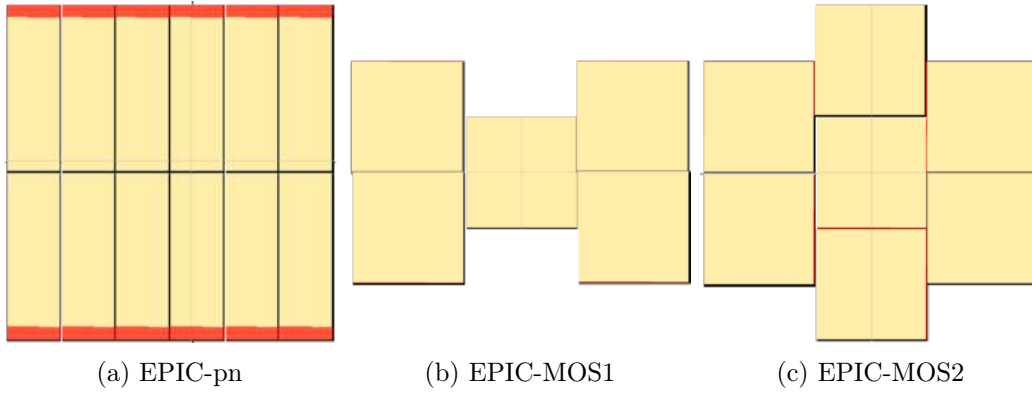


Figure 4.3: What the different instruments look like according to our XMLs files. The intersection of the grey lines gives the focal point, the red regions visualise to which end the pixels are shifted. Behind these regions are the respective readout amplifiers. Note that EPIC-MOS1 and EPIC-MOS2 have been rotated to match the coordinate system of EPIC-pn.

we argue that simulating n individual CCDs does not result in n -times longer simulation times compared to a single CCD. While it is true that there is a minor time penalty due to repeated initialisation and file reading, SIXTE is efficient enough to quickly discard photons that are out of the FOV and would never reach the CCD being simulated [48]. This efficiency minimises the performance impact when simulating multiple CCDs.

4.2.2 The Simulation Process

For the *extended sources*, such as galaxies and clusters, we utilise the TNG simulations introduced in section 2.2.2. Specifically, we use the TNG50-1, TNG100-1, and TNG300-1 simulations, as outlined in table 2.3, at a redshift of $z = 0.1$. For each simulation, the 400 most massive subhalos, based on gas mass, are retrieved. These subhalos are projected along the x , y , and z axes and viewed on two different scales, as detailed in table 4.1.

The choice of using both a close-up projection and one at a distance four times further allows for the capture of different spatial details from the same source, according to Sweere et al. [1], resulting in 2400 projections per simulation. From these projections, the X-ray photon intensity within the 0.5 keV to 15.0 keV energy range, corresponding to the energy spectrum of XMM-Newton, is computed. When creating the SIMPUTs, assigning a flux and a power model to these maps is necessary. The flux is uniformly sampled between 5 and 50 times the standard deviation of the background noise flux, σ_b , based on its mean, μ_b . The central part of the map, defined as a box covering 5% of the image width and height at the centre, is assigned this flux.

	TNG50-1	TNG100-1	TNG300-1
Scales	0.100, 0.400	0.400, 1.6	1.0, 4.0

Table 4.1: Scales in Mpc used for the various TNG simulations by Sweere et al. [1].

Unit	μ_b 10^{-12} W	σ_b 10^{-12} W	N_H 10^{22} cm^{-2}	Γ
Value	≈ 7.88	≈ 6.11	0.04	2

Table 4.2: Parameters for SIMPUT creation as suggested by Sweere et al. [1].

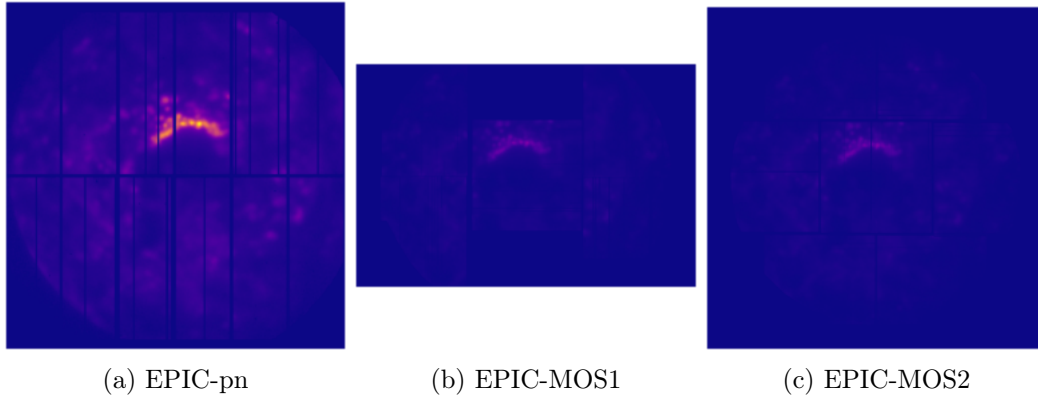


Figure 4.4: Resulting image from the instruments, in their respective resolution, for subhalo 542666 from TNG50-1 with an exposure time of 100 ks.

While galaxy clusters typically exhibit a thermal spectrum, the subhalos derived from different TNG scales are not necessarily galaxy clusters. Therefore, Sweere et al. [1] chose the absorbed power-law model for simplicity. The spectrum modelling was performed using XSPEC [131], applying the hydrogen column density N_H and the photon index Γ . The values of these four parameters can be found in table 4.2.

Subsequently, for each intensity map, five different SIMPUTs are generated. Each SIMPUT features a random zoom factor ranging between 1 and 2, as well as a random x and y shift, which are drawn from a standard distribution with a standard deviation of 5% of the image height and width. This process results in a total of 12 000 SIMPUTs for each TNG simulation. Figure 4.4 shows an example image for all instruments. Except for the colour, this image is exactly as it will be used during training. The colouring was added and the images have been resized to the same size for a more pleasurable viewing. This is the case for all image shown from now on.

The *point source component* of the observations predominantly consists of AGN, which are compact regions at the centres of galaxies exhibiting very high luminosity. Due to their compactness, AGN are typically only visible as point sources. Their expected number is calculated based on measurements of AGN number counts N and their relative fluxes S . This calculation is conducted for the energy band ranging from 0.5 keV to 2.0 keV. Since the authors of the original calculations did not publish their uncertainties, Sweere et al. [1] introduced an additional Poisson uncertainty to account for variability.

The absorption of AGN at different Galactic latitudes is simulated by shifting the $\frac{\log(N)}{\log(S)}$ distribution. A flux cut is applied at $1.5849 \times 10^{-15} \text{ erg cm}^{-2} \text{ s}^{-1}$, following the approach of Sweere et al. [1]. This ensures that any source fainter than this threshold will not be simulated. Furthermore, at least one observable AGN in each observation is required.

While AGN are distributed throughout the entire sky, their visibility is influenced by the Galactic plane, which partially absorbs X-rays from distant AGN. Consequently, their distribution is also dependent on Galactic latitude. This absorption is simulated by allowing the Galactic plane to fully absorb X-rays, with no absorption occurring when the line of sight is directed out of the Galactic plane. The absorption is linearly sampled between these extremes in logarithmic space, favouring observations outside the Galactic plane. An example image showcasing all instruments is presented in fig. 4.5.

All XMM-Newton observations contain noise, which necessitates the simulation of *background noise*. To achieve this, we utilise the prepared blank sky event list `pn_t_ff_g` [45], following the approach of Sweere et al. [1]. This event list allows us to extract the background noise across all energy levels. The background component is simulated using the

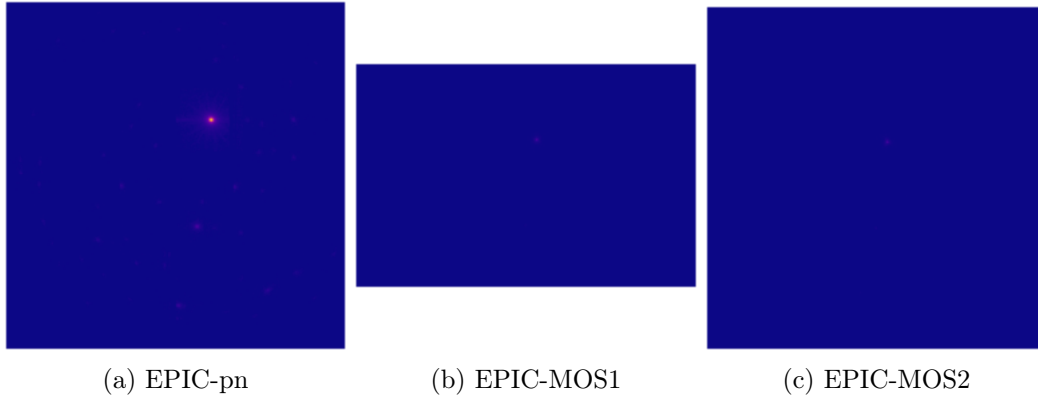


Figure 4.5: Resulting image from the instruments, in their respective resolution, for the same point source with an exposure time of 100 ks.

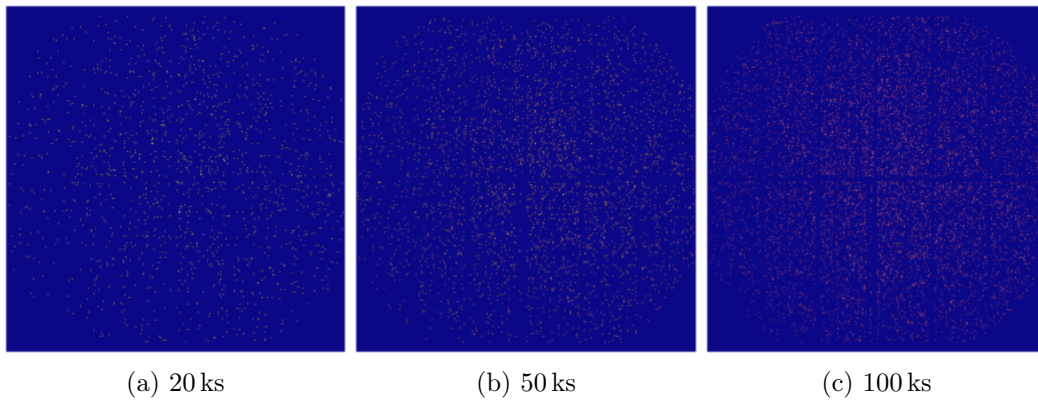


Figure 4.6: The background noise for EPIC-pn with different exposure times.

spectrum obtained from these Blank Sky event files. It is important to note that we only simulate the background for EPIC-pn, as our focus is specifically on enhancing images captured by this instrument. Figure 4.6 illustrates the background for EPIC-pn at various exposure times.

Once the SIMPUTs have been generated, SIXTE employs them to create an event list. This event list is filtered for a maximum pattern: four for EPIC-pn and twelve for EPIC-mos. We generate images from the filtered event list and apply a mask to eliminate out-of-FOV pixels (similar to the processing of real images) and remove dead pixels. Users can choose between the exposure and detector maps for this masking process. The exposure map encompasses the entire FOV, while the detector map focuses solely on the area where source detection will be conducted by SAS tasks. At a minimum, the exposure map should be applied to ensure that any insensitive detector areas are accurately removed.

In our simulations, we only enhance images from EPIC-pn at the higher resolution, while EPIC-mos detectors are maintained at their actual resolution. This decision is made because the pixel size of the EPIC-mos detectors is already smaller, and it helps to reduce simulation time.

4.3 Pre-Processing

To prepare the data for our models, we convert the raw counts from the images into counts per second by dividing each image by the corresponding exposure time. This process allows us to work with training data with varying exposure times while maintaining the integrity

of the input pixel intensity distribution. Unlike Sweere et al. [1], we do not clip the resulting pixel intensities to a maximum value, which helps preserve fine details in brighter regions.

Following this, the images are normalised to a range of values between 0 and 1. However, faint features may still be difficult to discern even after normalisation without proper data scaling. To enhance the visual clarity of these faint details, Sweere et al. [1] have explored several data scaling functions, including linear, square root, logarithmic, and hyperbolic arcsine stretch functions. Following the decision of Sweere et al. [1], we have decided to employ the square root function for scaling.

For the EPIC-mos images, we re-bin them with a factor of 4, as the pixel size of EPIC-MOS is $1''$ and that of EPIC-pn is $4''$. While it is true that the physical pixel size of the EPIC-MOS detectors is $1.1''$, it has to be noted that an SAS task exists, to extract EPIC-MOS images with our desired pixel size. This naive approach allows us to combine the EPIC-MOS images with the high-resolution images from EPIC-pn. This method is advantageous because EPIC-pn images are smaller, resulting in a reduced input size for the AI, leading to faster training.

When combining images from multiple detectors, we first add the images together before applying the normalisation process. This ensures a coherent data integration from different sources while maintaining the overall pixel intensity distribution.

4.4 Data Split

The final images generated from our simulations are divided into training, validation, and test sets. Only the training dataset is used to update the weights of the neural network, while the validation dataset serves to monitor the network's performance during training. The test dataset is reserved for the final evaluation of the network and is not exposed to the model during the training phase.

The splits are carefully designed to ensure that all spatial augmentations applied during the simulations are consistently contained within the same set. It's important to note that a specific source may appear multiple times across the sets, but with different projections and distances, which adds to the robustness of our training data.

For each component of the simulated dataset, extended sources, point sources, and background, we follow a distribution split of 80 % for training, 10 % for validation, and 10 % for testing, as recommended by Sweere et al. [1]. The real dataset is exclusively used for qualitative evaluation, where we have selected a small subset of images to ensure that the qualitative performance assessment is based on a consistent set of images. This approach allows us to effectively evaluate the generalisation and performance of our models on real-world data.

5 Methods

In this chapter, we delve into the models we will be using. The chapter is structured into three main sections, each addressing a critical aspect of the methodology employed in this work. Section 5.1 offers a comprehensive overview of the baseline models, including their architectures and key features. The subsequent sections summarise the individual models employed for SR, in section 5.2, and DN, in section 5.3, highlighting their roles and unique contributions to the overall framework. The final section of this chapter, section 5.4, describes the various configurations with which we train the models, as well as the loss functions utilised.

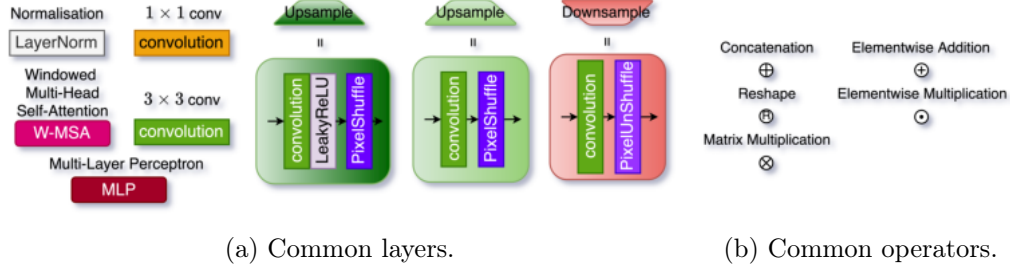


Figure 5.1: Some common components used throughout this chapter. The components are colour coded for easier recognition. The W-MSA layer has been introduced by Liang et al. [88] under the name MSA.

5.1 Baseline Models

The baseline models employed in this work have been introduced by Sweere et al. [1] in their 2022 paper. In their study, they have presented two distinct models: one specifically designed for DN and another for the joint task of SR and DN. The DN model focuses on reducing noise in XMM-Newton images, preserving crucial details while enhancing image clarity. The second model extends this approach by simultaneously performing SR, thereby improving both the images' spatial resolution and signal-to-noise ratio. This section thoroughly examines these baseline models, discussing their architectures, key innovations, and performance metrics. These models serve as the foundation for the current study, providing a benchmark for evaluating additional models used in our work.

They initially chose the ESRGAN model [85] for its proven ability to enhance image resolution. The model features a stacked RRDB generator and a deep CNN discriminator.

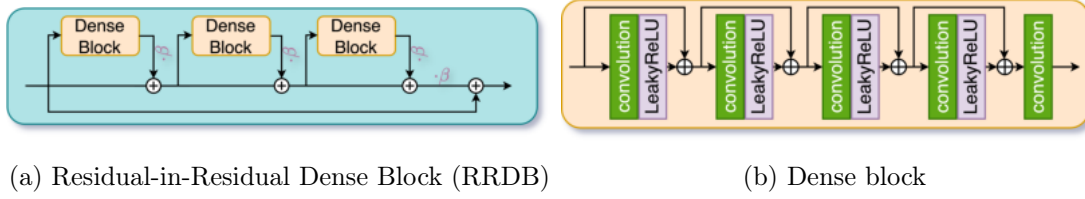


Figure 5.2: The RRDB and its dense block. Adapted from Wang et al. [85].

However, like all GAN-based algorithms, the ESRGAN model is susceptible to producing non-existent features in the output, known as hallucinations. The baseline models employed in this work have been introduced by Sweere et al. [1] in their 2022 paper. They presented two distinct models: one specifically designed for DN and another for the joint task of SR and DN. The DN model focuses on reducing noise in XMM-Newton images, preserving crucial details while enhancing image clarity. The second model extends this approach by simultaneously performing SR, thereby improving both the images' spatial resolution and signal-to-noise ratio. This section thoroughly examines these baseline models, discussing their architectures, key innovations, and performance metrics. These models serve as the foundation for the current study, providing a benchmark for evaluating additional models used in our work.

Sweere et al. [1] initially chose the ESRGAN model [85] for its proven ability to enhance image resolution. The model features a stacked RRDB generator and a deep CNN discriminator. However, like all GAN-based algorithms, the ESRGAN model is susceptible to producing non-existent features in the output, known as hallucinations, as pointed out by Sweere et al. [1]. These occurrences stem from the discriminator network, which influences the generator output to mimic the images in the training dataset closely. When the input image lacks sufficient information to generate a high-resolution equivalent, the model may hallucinate intricate features to create a visually similar output image. Such occurrences can have significant implications in astronomy.

To achieve more robust reconstructions, albeit at the expense of generating less visually similar images, Sweere et al. [1] removed the adversarial component. As a result, the primary model is based solely on the RRDB generator used in ESRGAN, making the architecture no longer operate as a GAN. This approach yields more reliable outputs; however, it can only generate sharp reconstructions in areas where the model has confidently learned to generate features, such as high-signal-to-noise point sources. Sweere et al. [1] emphasise that the model does not provide a confidence level for the generated features. Regions of low confidence, such as background noise, can result in blurry reconstructions, and the output image may not visually resemble the target image. The confidence level varies for each image, depending on the features present in the input image.

According to its creators, Wang et al. [85], the RRDB is based on the idea that increasing the number of layers and connections can enhance performance. It employs a deeper and more complex structure compared to the original residual block in SRGAN [84], which ESRGAN is based on. The RRDB features a residual-in-residual structure, using residual learning at different levels, as depicted in fig. 5.2. While a similar network structure is proposed in Zhang et al. [132], which also utilises a multi-level residual network, the RRDB differs by incorporating a dense block [133] in the main path. This approach increases the network capacity by leveraging dense connections. Additionally, several techniques are employed to facilitate training a very deep network, including residual scaling to prevent instability by scaling down the residuals by β , where $0 < \beta < 1$, before adding them to the main path, and using smaller initialisation parameters, which have been empirically found to make training the residual architecture easier. For further insight, refer to [85] and the

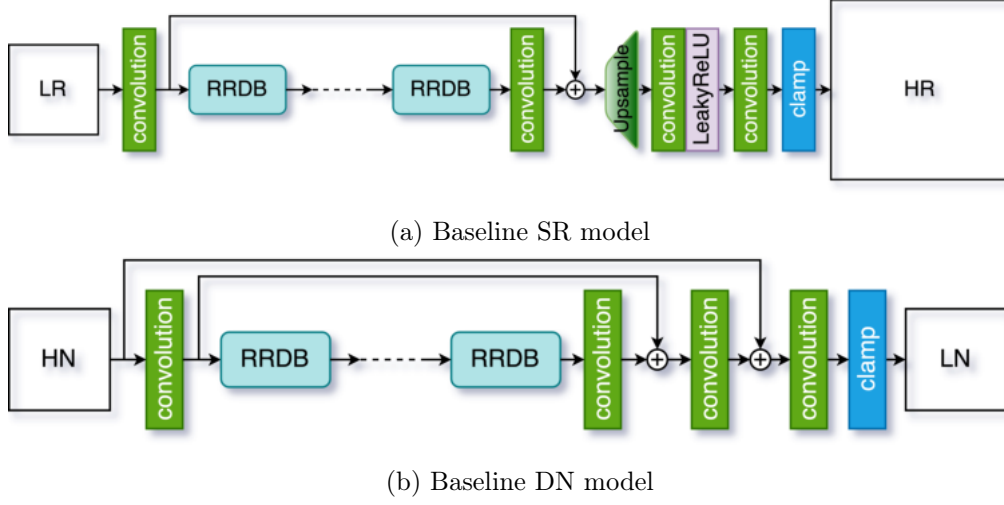


Figure 5.3: The models used by Sweere et al. [1]. Adapted from their work.

supplementary material. LeakyReLU [134] is a ReLU variant with a non-zero gradient over its entire domain.

The *SR model* is nearly identical to the RRDB generator and is given fig. 5.3a. The distinction is that Sweere et al. [1] utilise PixelShuffle upsampling [83] instead of the nearest-neighbour interpolation upsampling and that the last convolutional layer is followed by clamping the pixels values between 0 and 1. Pixel shuffle involves more connections and avoids interpolating the upsampled image, which is expected to enhance the quantitative details on smaller scales. Furthermore, they have set $\beta = 0.2$.

The same architecture is utilised in the *DN model*, as pointed out by Sweere et al. [1], but the upsampling layer is omitted. Furthermore, additional skip connections are incorporated as inspired by Zhang et al. [135]. This allows the model to capture smaller image features effectively and enhances training efficiency by focusing on learning features to suppress and produce the denoised output image. Figure 5.3b shows the architecture of this model.

As the output images have values ranging from 0.0 to 1.0, ensuring that the initial pass through the model yields values within this range is crucial, according to Sweere et al. [1]. If this does not occur, for instance, if the values are all negative, everything will be clipped to 0, resulting in no usable gradients for back-propagation. Consequently, the model will not train effectively. To mitigate this, the initial weights in the final convolution layer are adjusted to be slightly more positive. Generally, the weights are initialised using a random normal distribution, with the standard deviation being based on the size of the convolution layer:

$$\sigma = \frac{1}{\sqrt{\text{layer size}}} \quad (5.1)$$

To prevent the initial forward passes from falling outside the image range, the weights in the last convolution layer are initialised to be uniformly distributed from $[-\sigma, 1.01 \cdot \sigma]$, ensuring that the weights are slightly more positive.

5.2 Super-Resolution Model

When utilising Transformer-based SR models for inference across diverse datasets, Hsu et al. [2] identified a consistent trend: the intensity distribution of the feature maps undergoes significant changes with increasing network depth. This evolution reflects the model’s learning of spatial information and attention intensity. Nevertheless, towards the later

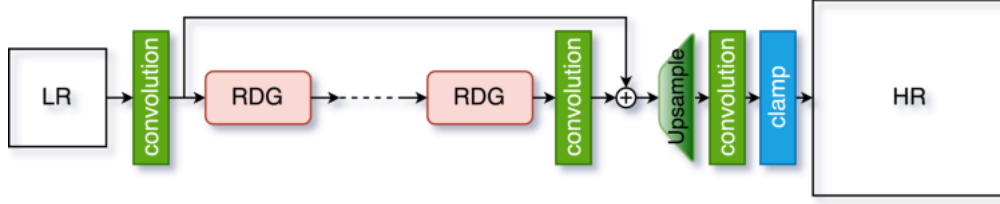


Figure 5.4: The architecture of DRCT. Adapted from Hsu et al. [2].

stages of the network, there is typically a sharp decline in intensity, resulting in a narrower range. This pattern indicates that such abrupt changes may lead to a loss of spatial information, suggesting the existence of an information bottleneck.

As per the information bottleneck principle [136], the input data X may lead to information loss as it traverses through consecutive layers, potentially resulting in gradient vanishing during back-propagation for parameter fitting and Y prediction, as depicted in the equation:

$$I(X, X) \geq I(Y, X) \geq I(Y, f_{\theta}X) \geq I(X, g_{\phi}(f_{\theta}(X))). \quad (5.2)$$

Here, I represents mutual information, f and g denote transformation functions, and θ and ϕ represent parameters for f and g respectively. In the context of deep neural networks, f_{θ} and g_{ϕ} correspond to two successive layers. This equation suggests that as the network's layer count increases, the likelihood of information loss in the information flow also increases.

Hsu et al. [2] posit that network architectures based on SwinIR [88], while effectively expanding the receptive fields using a shift-window attention mechanism [88] to address the limited receptive fields in CNNs, are susceptible to gradient bottlenecks as the network depth increases, which effectively constrains the model's performance and potential. To tackle the problem of spatial information loss caused by a growing number of network layers, Hsu et al. [2] have developed the Dense-Residual-Connected Transformer (DRCT), which is shown in fig. 5.4. This framework aims to stabilise the forward-propagation process and alleviate information bottlenecks. The Swin-Dense-Residual-Connected Block (SDRCB), proposed in their work and shown in fig. 5.5c, integrates Swin Transformer Layers (STLs), as shown in fig. 5.5b, and transition layers into each Residual Deep feature extraction Group (RDG) to achieve this goal, as can be seen in fig. 5.5a.

In the development of RDG, Hsu et al. [2] drew inspiration from ESRGAN [85] and RDN [118], utilising a Residual-Dense Block (RDB) as the fundamental unit for SR. Reusing feature maps results in an expanded receptive field within the feed-forward mechanism of RDG. Additionally, employing several SDRCB in RDG enhances the ability to fuse information across multiple scales, enabling more comprehensive feature extraction. RDG promotes the flow of information within the residual group, capturing local features and spatial information group by group. To capture long-range dependencies effectively, Hsu et al. [2] incorporate the shifting window self-attention mechanism of STLs into their approach, which allows adaptive adjustments of the receptive fields, enhancing ESRGAN [85] by focusing on multi-level spatial information. By leveraging STL, the model's focus can be dynamically shifted based on the global content of the input, leading to a more precise and efficient feature extraction process. This mechanism ensures that global details are preserved even with increasing network depth, effectively expanding and improving the receptive field without compromise. Through the integration of STLs with dense-residual connections, this architecture benefits from a broad receptive field and the ability to pinpoint the most relevant information, thereby enhancing the model's performance in

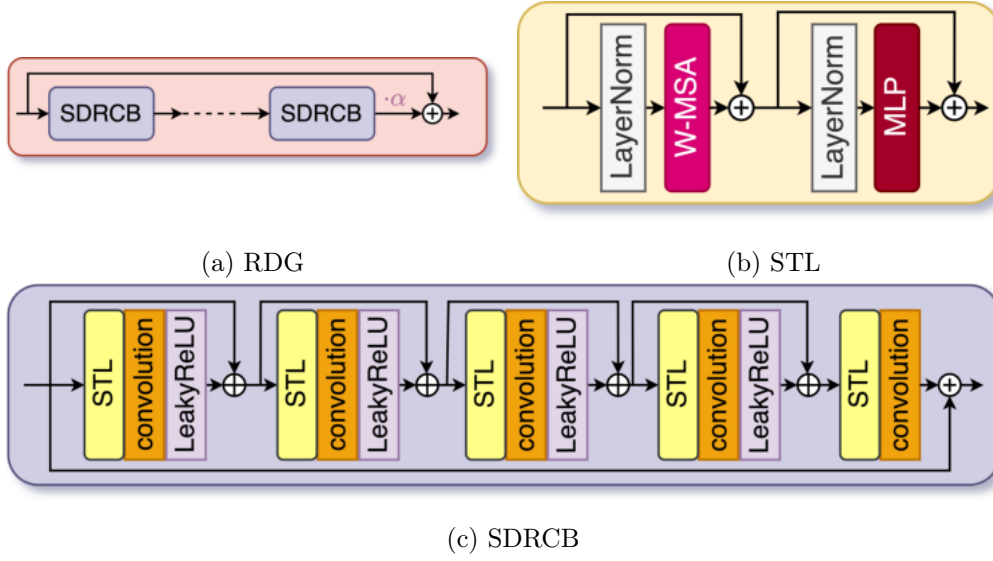


Figure 5.5: The Residual Deep feature extraction Group (RDG), consisting of Swin-Dense-Residual-Connected Blocks (SDRCBs), which in turn contains Swin Transformer Layers (STLs). Adopted from Hsu et al. [2].

SR tasks that demand detailed and context-aware processing, as highlighted by Hsu et al. [2].

The last SDRCB output is multiplied by a residual scaling factor α , which is set to 0.2 for stabilising the training process, according to Hsu et al. [2]. Furthermore, the negative slope of LeakyReLU is also set to 0.2. The last convolution in the deep feature extraction is used to fuse various features with different levels [137] adaptively. The HR image is then reconstructed by aggregating shallow and deep features through two convolutions and a PixelShuffle [83] layer in between.

5.3 Denoising Model

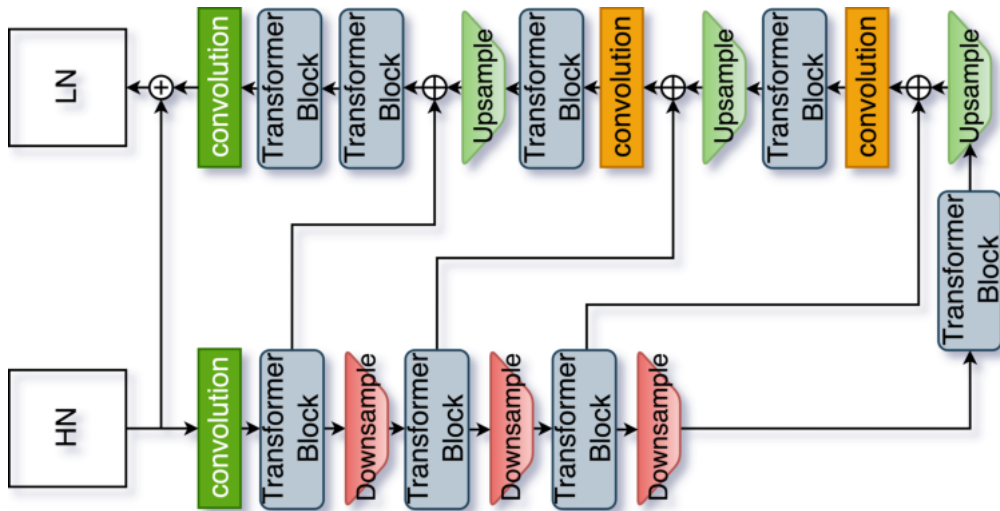


Figure 5.6: Architecture of Restormer. Adapted from Zamir et al. [3].

Zamir et al. [3] introduced Restormer, an efficient transformer designed specifically for image restoration. This model is adept at capturing global connectivity and is well-suited

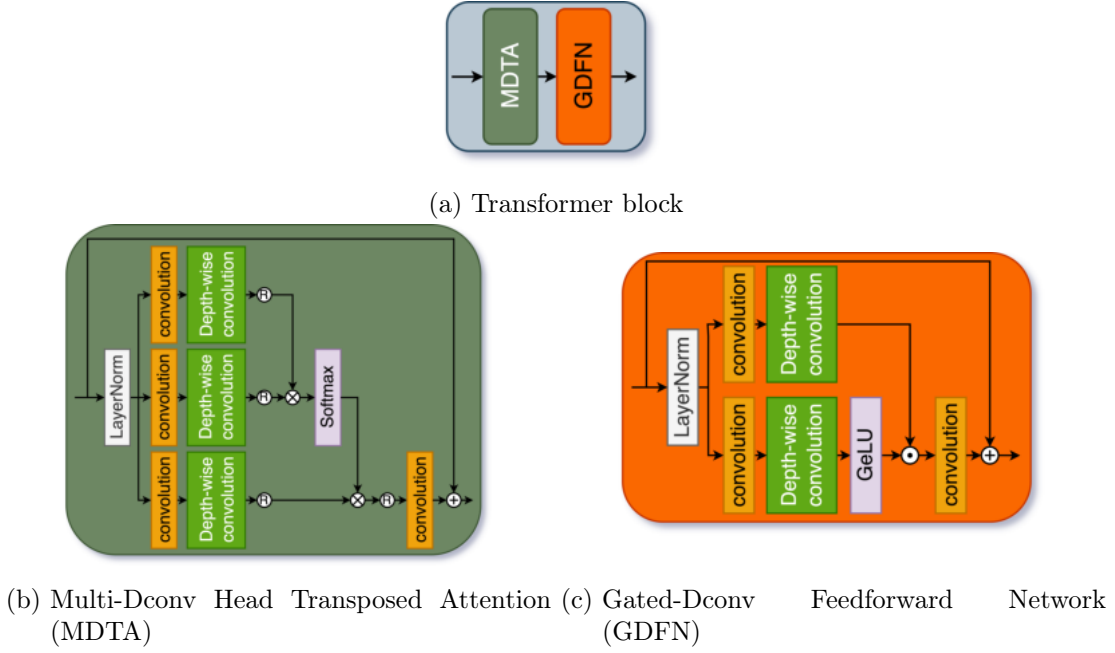


Figure 5.7: The transformer block and its components. Adapted from Zamir et al. [3].

for processing large images. The authors incorporated a Multi-Dconv Head Transposed Attention (MDTA) block, illustrated in fig. 5.7b, which replaces the traditional multi-head self-attention mechanism [68]. This block features linear complexity, allowing it to efficiently aggregate both local and non-local pixel interactions, making it ideal for high-resolution image processing. Additionally, they developed the Gated-Dconv Feedforward Network (GDFN), depicted in fig. 5.7c, which is a feed-forward network with a gating mechanism [138]. This mechanism selectively suppresses less informative features, ensuring that only the most valuable information progresses through the network hierarchy.

When processing a degraded image, Restormer first applies a convolution to extract low-level feature embeddings, as described in Zamir et al. [3]. These embeddings are then processed through a four-level symmetric encoder-decoder architecture to transform them into deep features. Each level of the encoder-decoder contains multiple transformer blocks, with the number of blocks gradually increasing from top to bottom to enhance efficiency. One such block is shown in fig. 5.7a. The encoder progressively reduces spatial size while expanding channel capacity, starting from the high-resolution input, while the decoder takes low-resolution latent features and incrementally reconstructs the high-resolution representations.

To facilitate feature downsampling and upsampling, Zamir et al. [3] employ PixelUnshuffle and PixelShuffle operations [83]. Additionally, skip connections [116] are used to link encoder features with decoder features, aiding the recovery process. Following the concatenation of these features, a 1×1 convolution reduces the channel count by half at all levels except for the top one. At level 1, the transformer blocks aggregate low-level features from the encoder with high-level features from the decoder, which helps preserve fine structural and textural details in the restored images.

The deep features undergo further enhancement during a refinement stage, which operates at a high spatial resolution. These design choices collectively contribute to significant quality improvements, as detailed by Zamir et al. [3]. Finally, a convolution layer is applied to the refined features to generate the residual image, which is then added to the degraded image to obtain the restored output.

Zamir et al. [3] highlight that the primary computational overhead in transformers arises from the self-attention layer. In conventional self-attention [68], both the time and memory complexity of the key-query dot-product interaction scale quadratically with the spatial resolution of the input, resulting in a complexity of $\mathcal{O}(W^2H^2)$ for images of size $W \times H$ pixels. The MDTA mitigates this issue by applying self-attention across channels rather than the spatial dimension, computing cross-covariance across channels to produce an attention map that implicitly encodes global context. Furthermore, the MDTA incorporates local context mixing prior to the feature covariance computation. This is achieved through pixel-wise aggregation of cross-channel context using convolutions and channel-wise aggregation of local context using efficient depth-wise convolutions.

Zamir et al. [3] introduce two fundamental modifications to the feed-forward network to enhance representation. Firstly, the gating mechanism, which is the element-wise product of two parallel paths of linear transformation layers, with one path activated using the GELU non-linearity [139]. The gating mechanism in GDFN regulates which complementary features should be propagated forward. It enables subsequent layers in the network hierarchy to concentrate on more refined image attributes, producing high-quality outputs. Secondly, depth-wise convolutions are included in GDFN to capture information from neighbouring pixel positions in space. This is valuable for learning local image structure for effective restoration. GDFN serves a distinct purpose compared to MDTA, which focuses on enriching features with contextual information. The architecture of GDFN is depicted in fig. 5.7c.

5.4 Training Configuration

As mentioned in chapter 4 we use the simulated dataset for training, validation and testing. The dataset of real observation are only used for visual evaluation. For the *DN* task, we use images with a 20 ks exposure, intentionally adding background noise and random AGN. The target images, however, are cleaner, with a longer 50 ks exposure, free from background noise but containing the same AGN. This was the choice for both the baseline model as well as for Restormer. For the *SR* task we have tried different combinations, but all have the same exposure for the LR image (20 ks) and for the HR image (100 ks) with a resolution multiplier of two, i.e., double the image size, half the pixel size, and half the PSF. Here we experiment how the addition of background to the LR and EPIC-MOS to the HR images affect the results. A summary can be found in tables 5.1 and 5.2. The values for AGN, Background (BKG), and EPIC-MOS give if the component is used (1) or omitted (0). If there is just one given value than that value is applicable for the whole column.

Input				Target			
Exposure	AGN	BKG	EPIC-MOS	Exposure	AGN	BKG	EPIC-MOS
20 ks	1	1	0	50 ks	1	0	0

Table 5.1: Our training configuration for DN.

Input				Target			
Exposure	AGN	BKG	EPIC-MOS	Exposure	AGN	BKG	EPIC-MOS
20 ks	1	1	0	100 ks	1	0	0
		0					0
		1					1

Table 5.2: Our training configuration for SR.

For all our models we use the Adam optimiser [140] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate is set to 0.0001 for the baseline models as well as Restormer, while DRCT

uses a learning rate of 0.0002. We trained the baseline models on all configurations for 50 epochs. The Restormer model was only trained on the single DN task for 10 epochs, and DRCT has been only trained on the first of the three SR configurations for 3. These choices for the transformer models were made due to time and hardware constraints. We have kept the configuration of the baseline models as described by Sweere et al. [1]. For DRCT and Restormer we have only reduced the patch size to 32 and 24, respectively, but otherwise kept the configuration as described in their introducing paper. This choice has also been done because of hardware constraints.

Based on the model’s performance, the image quality metrics and visual inspections of the generated images, Sweere et al. [1] opted to train their models using a combination of PSNR and MS-SSIM. They intentionally excluded MSE due to its sensitivity to outliers and extreme values, which can negatively impact performance, particularly when working with AGN observations. MS-SSIM requires fine-tuning of parameters to address our specific problem. Sweere et al. [1] empirically determined that the following parameters worked well:

kernel size	σ	$k1$	$k2$	weights
13	2.5	0.01	0.05	0.0448, 0.2856, 0.3001, 0.2363 and 0.1333

After their hyper-parameter search, they settled on the following normalised loss function:

$$L_b(\mathbf{y}, \hat{\mathbf{y}}) = 5.43 - 0.0609 \cdot \text{PSNR}(\mathbf{y}, \hat{\mathbf{y}}) - 1.51 \cdot \text{MS-SSIM}(\mathbf{y}, \hat{\mathbf{y}}) \quad (5.3)$$

We refer to their paper for more detailed information on Sweere et al.’s [1] hyper-parameter search process, the considered loss functions, and the normalisation procedure.

6 Evaluation

This chapter evaluates our models through both quantitative and qualitative approaches. We examine a variety of performance metrics, such as MAE, PSNR, Poisson, SSIM, MS-SSIM, FSIM, and HaarPSI, which were previously introduced in section 2.7.1. The results are organised into two sections: quantitative results, where we assess the models’ numerical performance using these metrics, and qualitative results, where we visually evaluate the models’ effectiveness. Section 6.1 and section 6.2 present the outcomes for SR and DN, respectively.

6.1 Super-Resolution

The quantitative performance was evaluated across seven different test settings, with the results summarised in tables 6.2a to 6.2e. These test settings are defined in table 6.1.

	AGN	BKG	EPIC-MOS	Dataset
1	1	1	0	Our newly simulated dataset
2	1	1	1	Our newly simulated dataset
3	1	0	0	Our newly simulated dataset
4	1	0	1	Our newly simulated dataset
5	1	1	0	The simulated dataset by Sweere et al. [1]
6	1	0	0	The simulated dataset by Sweere et al. [1]
7	0	0	0	The real dataset by Sweere et al. [1]

Table 6.1: The different test settings for our SR models.

Focusing on the cases where the baseline model was trained on our dataset, as presented in tables 6.2a to 6.2c, we observe that the best results were obtained in settings 1 to 4. This is expected, as the model was trained on our data in these instances. Moreover, the top and second-best performances are almost exclusively found in test settings 1 and 3, both of which omit EPIC-MOS. Even the model trained on setting 2 performs best on these same settings, which makes sense given that these settings introduce the least additional information, with the background contributing far less than the EPIC-MOS images.

Interestingly, the model trained on the dataset from Sweere et al. [1] also outperforms on our dataset in most cases, as shown in table 6.2d. It is important to note that the values in column 5 of table 6.2d are based on clipped values, as described in their paper. We

argue that this clipping accounts for the observed difference in metric values, and that in all other cases, the performance on their simulated dataset is lower. Clipping reduces details in bright areas and background regions near the clipping threshold, which simplifies the model’s learning process. The impact of removing this clipping in setting „5“ has not been evaluated. Overall, these metrics suggest that our newly simulated dataset enhances performance, and it appears that the addition of EPIC-MOS (as shown in column „2“) does not improve performance on the simulated dataset. Furthermore, we note that we were not able to reproduce Sweere et al.’s [1] results, since we have achieved worse SR performance.

The performance of DRCT, as shown in table 6.2e, demonstrates that even after just three epochs, it achieves results comparable to the baseline model. However, similar to the baseline, DRCT performs better on our dataset. Its lower performance on the dataset by Sweere et al. [1] is in line with the baseline models tested on that dataset, with the exception of the baseline model trained on the clipped images by Sweere et al. [1] This further reinforces the concern that clipping the images may artificially inflate performance by excessively removing important details along with the noise.

Shifting our focus to the baseline model’s performance on the real dataset, presented in table 6.3a, we see that for nearly all metrics, the model trained on our dataset achieves either the best or second-best results. The only exception is FSIM, where the model trained on Sweere et al.’s [1] simulated dataset performs the best. Unlike the results on simulated datasets, there is no clear indication that specific training settings offer a distinct advantage here. Additionally, as shown in table 6.3b, DRCT surpasses the baseline model in almost all metrics. It’s important to note that DRCT was only trained for three epochs due to time and hardware limitations, meaning further training is needed to confirm these results. For the evaluation on the real dataset, we used the nearest neighbour method to upscale the real observations, as no HR images are available in this dataset.

	1	2	3	4	5	6
MAE	0.003 24	0.003 74	0.003 31	0.003 78	0.008 40	0.008 38
PSNR	37.606 04	36.250 91	37.308 95	36.200 19	31.112 22	30.912 99
Poisson	0.021 67	0.025 97	0.022 07	0.026 35	0.050 16	0.051 65
SSIM	0.945 97	0.934 75	0.943 54	0.931 89	0.879 01	0.864 42
MS-SSIM	0.976 40	0.969 09	0.974 72	0.967 33	0.955 75	0.946 37
FSIM	0.896 16	0.898 88	0.883 53	0.887 27	0.884 63	0.861 69
HaarPSI	0.747 33	0.741 08	0.746 94	0.739 90	0.624 74	0.621 41

(a) SR performance of the baseline model trained on the first setting

	1	2	3	4	5	6
MAE	0.003 37	0.003 50	0.003 33	0.003 56	0.008 11	0.008 10
PSNR	37.643 78	37.302 70	37.683 48	37.138 71	32.000 50	31.707 97
Poisson	0.021 31	0.025 22	0.021 36	0.025 75	0.049 17	0.051 26
SSIM	0.951 86	0.945 28	0.951 13	0.942 66	0.891 27	0.873 45
MS-SSIM	0.980 48	0.978 24	0.979 74	0.976 73	0.965 75	0.957 17
FSIM	0.872 14	0.878 58	0.863 52	0.868 99	0.884 34	0.853 44
HaarPSI	0.752 16	0.769 14	0.755 36	0.768 05	0.646 81	0.640 20

(b) SR performance of the baseline model trained on the second setting

	1	2	3	4	5	6
MAE	0.003 24	0.003 83	0.003 19	0.003 75	0.008 52	0.008 42
PSNR	37.666 07	36.156 91	37.734 60	36.251 16	31.029 03	30.865 70
Poisson	0.021 58	0.026 37	0.021 42	0.026 05	0.050 57	0.051 51
SSIM	0.946 81	0.933 78	0.946 43	0.934 54	0.878 57	0.865 50
MS-SSIM	0.976 98	0.969 40	0.976 30	0.968 91	0.956 25	0.947 84
FSIM	0.896 50	0.898 58	0.882 85	0.886 07	0.882 34	0.859 96
HaarPSI	0.746 60	0.738 38	0.749 52	0.740 00	0.617 12	0.615 94

(c) SR performance of the baseline model trained on the third setting

	1	2	3	4	5	6
MAE	0.004 45	0.004 12	0.004 36	0.004 07	0.006 36	0.008 42
PSNR	34.933 37	35.987 18	35.085 09	36.002 24	35.787 91	31.702 47
Poisson	0.023 23	0.027 09	0.023 33	0.026 84	0.048 62	0.048 40
SSIM	0.943 57	0.942 96	0.943 33	0.943 28	0.907 37	0.885 10
MS-SSIM	0.966 19	0.970 91	0.966 41	0.970 61	0.974 00	0.954 39
FSIM	0.834 22	0.839 37	0.830 63	0.836 05	0.866 48	0.840 02
HaarPSI	0.596 48	0.636 72	0.597 97	0.637 02	0.664 81	0.617 37

(d) SR performance of the baseline model trained on the fifth setting. Note: The performance in column „5“ is the performance with clipped images. This corresponds to the configuration of Sweere et al. [1] and acts as our try to reproduce their SR performance.

	1	5	6
MAE	0.003 47	0.009 47	0.009 60
PSNR	36.882 34	30.109 26	29.924 28
Poisson	0.022 89	0.054 27	0.056 70
SSIM	0.935 58	0.856 10	0.838 51
MS-SSIM	0.964 37	0.938 33	0.927 28
FSIM	0.914 33	0.887 21	0.876 11
HaarPSI	0.724 10	0.579 99	0.576 14

(e) SR performance of DCRT trained on the first setting.

Table 6.2: SR performance on the simulated datasets of models trained on the various settings. The **best** and **second best** metric values are given in their respective colour.

	1	2	3	5
MAE	0.001 33	0.001 34	0.001 35	0.001 61
PSNR	40.765 25	40.968 48	40.370 28	38.109 70
Poisson	0.007 42	0.007 30	0.007 43	0.008 45
SSIM	0.964 89	0.965 52	0.965 11	0.956 98
MS-SSIM	0.931 05	0.929 02	0.934 30	0.933 11
FSIM	0.811 58	0.817 62	0.817 39	0.913 43
HaarPSI	0.304 72	0.303 18	0.297 93	0.204 00

(a) SR performance of the baseline model evaluated on the real dataset

	1
MAE	0.001 20
PSNR	41.875 47
Poisson	0.007 25
SSIM	0.965 62
MS-SSIM	0.911 92
FSIM	0.782 78
HaarPSI	0.331 68

(b) SR performance of DRCT evaluated on the real dataset

Table 6.3: SR performance of the models on the real dataset trained on the training settings given in the table header. The **best** and **second best** metric values are given in their respective colour.

Quantitative performance is not the only objective of SR models, and high scores do not necessarily translate to good qualitative results. In fig. 6.1 we show the predicted images from the models alongside the corresponding input, fig. 6.1a, and target image, fig. 6.1b, from our simulated dataset. It is important to note that, while the images appear to be the same size, they actually contain different pixel counts. This decision was made to streamline the visualisation process.

Sweere et al.’s [1] model produces promising results overall but exhibits a loss of detail near the edges of the FOV and in fainter regions. As previously discussed, this is likely related to the clipping of the images. In contrast, our baseline models show a marked improvement in preserving details across the entire image. Notably, it retains the structural information near the edges of the FOV, offering a more comprehensive reconstruction of the simulated data. On the other hand, DRCT, depicted in fig. 6.1g, appears to preserve information across all regions, although it requires considerably longer training. Nevertheless, its performance after just three epochs is highly promising.

Similar conclusions can be drawn regarding performance on real images. Figure 6.2 displays an input image in fig. 6.2a, alongside the predicted outputs from our baseline models in figs. 6.2b to 6.2d, Sweere et al.’s model [1] in fig. 6.2e, and DRCT in fig. 6.2f. Our baseline models retain significantly more information, even in the fainter regions. In comparison, Sweere et al.’s model retains only the details of the brightest region, while DRCT fails to produce satisfactory results.

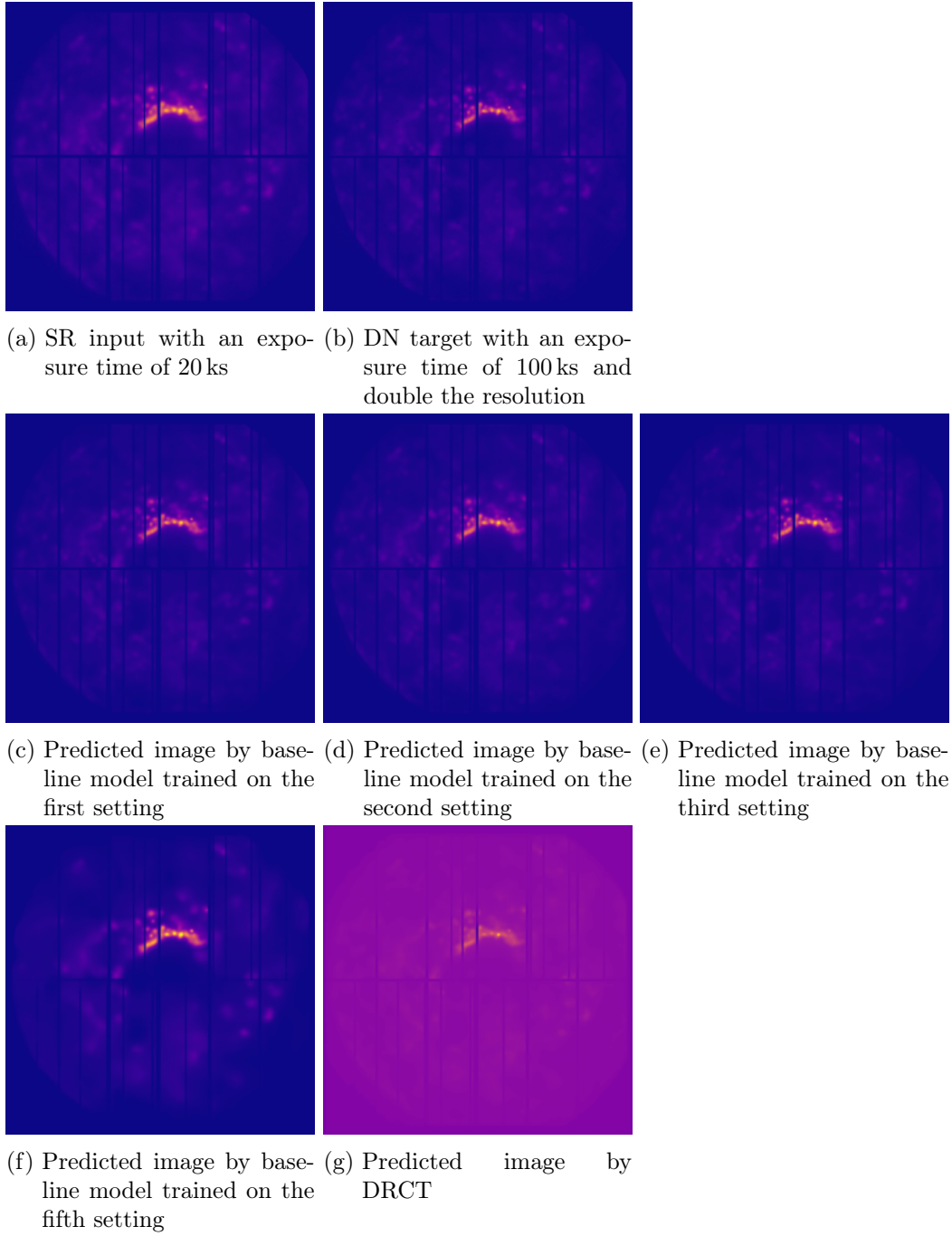
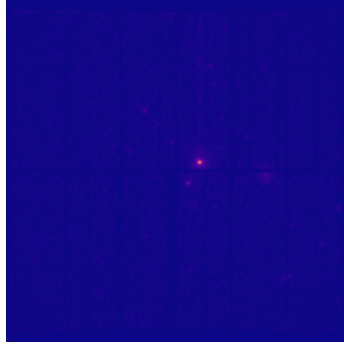
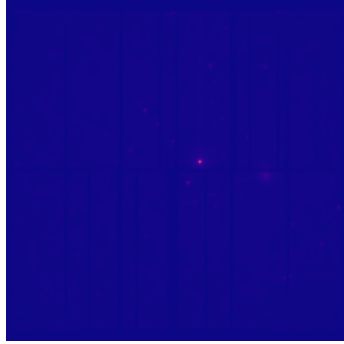


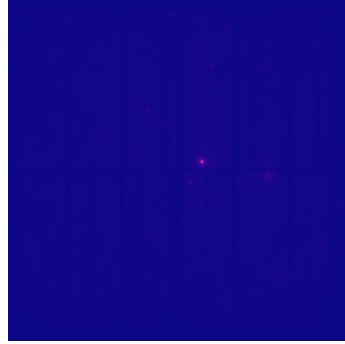
Figure 6.1: SR performance of the models on our simulated dataset



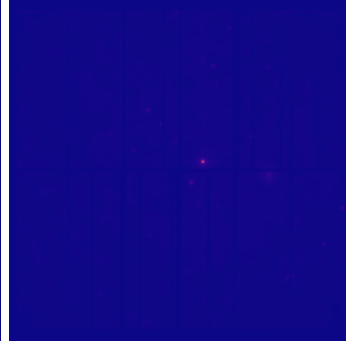
(a) SR input with an exposure time of 20 ks



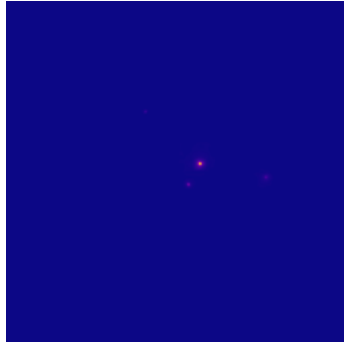
(b) Predicted image by baseline model trained on the first setting



(c) Predicted image by baseline model trained on the second setting



(d) Predicted image by baseline model trained on the third setting



(e) Predicted image by baseline model trained on the fifth setting



(f) Predicted image by DRCT

Figure 6.2: SR performance of the models on the real dataset

6.2 Denoising

Quantitative performance was evaluated exclusively on test settings 1, 5, and 7, as defined in table 6.1. The results are summarised in tables 6.4a and 6.4c.

	1	5	7
MAE	0.005 06	0.008 71	0.004 96
PSNR	35.230 16	32.393 12	36.118 37
Poisson	0.038 68	0.064 90	0.036 33
SSIM	0.931 51	0.913 27	0.914 88
MS-SSIM	0.983 72	0.977 21	0.951 38
FSIM	0.917 76	0.898 92	0.958 47
HaarPSI	0.817 78	0.773 72	0.876 43

(a) DN performance of the baseline model trained on the first setting

	1	5	7
MAE	0.005 32	0.004 13	0.004 86
PSNR	34.655 81	39.153 11	37.070 49
Poisson	0.041 58	0.044 11	0.041 04
SSIM	0.908 66	0.958 55	0.878 35
MS-SSIM	0.967 57	0.988 61	0.887 91
FSIM	0.850 10	0.919 48	0.874 65
HaarPSI	0.812 66	0.834 97	0.882 66

(b) DN performance of the baseline model trained on the fifth setting. Note: The performance in column „5“ is the performance with clipped images. This corresponds to the configuration of Sweere et al. [1] and acts as our try to reproduce their DN performance.

	1	5	7
MAE	0.005 02	0.008 64	0.004 45
PSNR	35.319 50	32.488 31	36.847 84
Poisson	0.038 74	0.064 88	0.035 25
SSIM	0.932 95	0.913 81	0.927 12
MS-SSIM	0.984 42	0.979 49	0.150 46
FSIM	0.922 40	0.898 72	0.965 96
HaarPSI	0.826 32	0.774 55	0.864 42

(c) DN performance of Restormer trained on the first setting

Table 6.4: DN performance of the models trained either on the first or the fifth setting. The **best** and **second best** metric values are given in their respective colour.

Our DN models show a performance trend similar to that of our SR models, with the models trained on our dataset outperforming the one from Sweere et al. [1]. In fact, they even perform better on the real dataset than on the simulated one. The model trained by Sweere et al. [1], whose performance is detailed in table 6.4b, exhibits behaviour similar to their SR model, achieving better results than our models. However, the same critique applies here: their use of clipping enhances performance, but we argue that this improvement is due to the excessive removal of noise. The Restormer model performs similarly to the baseline models trained on our dataset, suggesting that it requires significantly more training beyond the initial ten epochs to reach its full potential. Furthermore, we note that we were not able to reproduce Sweere et al.’s [1] results, since they have achieved better DN performance.

Figure 6.3 shows the predicted images from the models alongside the corresponding input, fig. 6.3a, and target image, fig. 6.3b, from our simulated dataset. The most striking observation is that Restormer, fig. 6.3e, fails to produce satisfactory results, likely due to its

brief training duration. When comparing our baseline model, fig. 6.3c, with that of Sweere et al. [1], fig. 6.3d, it becomes evident that our model preserves more faint features, while theirs tends to overlook them.

The visual performance on real images, as shown in fig. 6.4, reveals a similar outcome for the baseline models, though Restormer, fig. 6.4d, continues to fall short in delivering satisfactory results. Sweere et al.’s [1] baseline model, fig. 6.4c, loses nearly all detail, retaining only the bright central region. In contrast, our baseline model, fig. 6.3c, produces a visually more pleasing result, with significantly less noise.

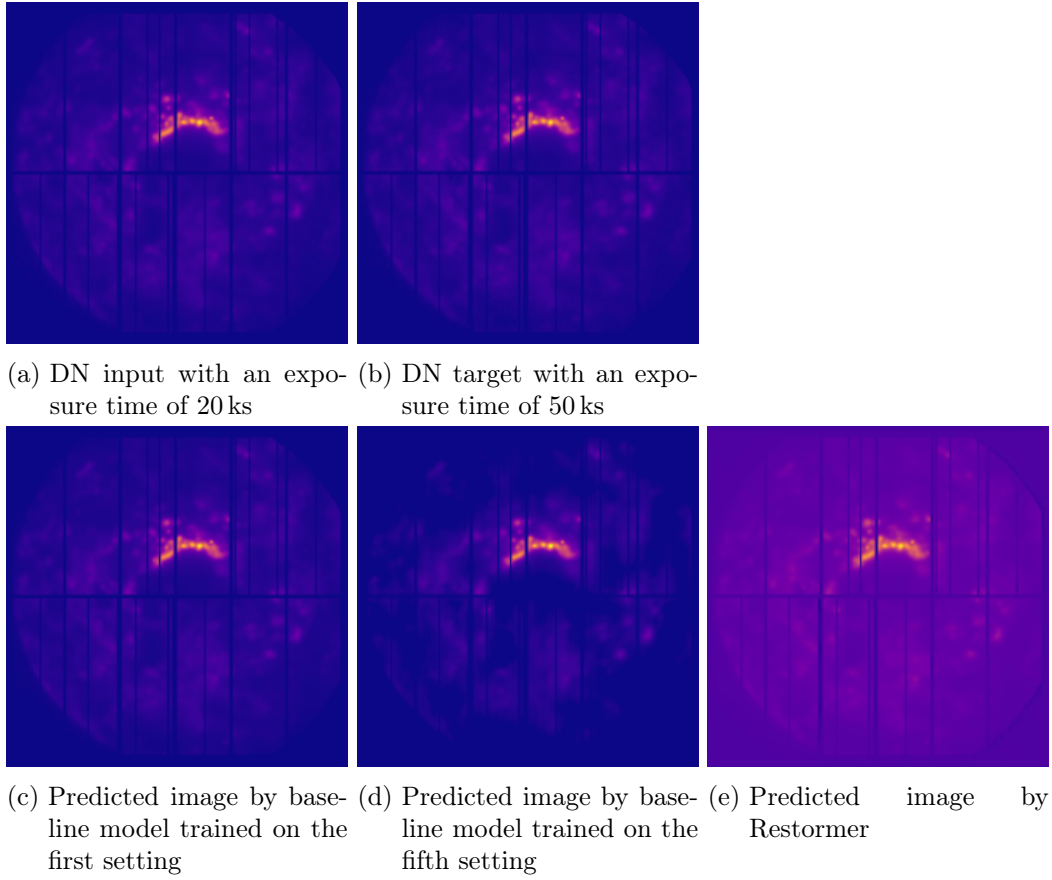


Figure 6.3: DN performance of the models on our simulated dataset

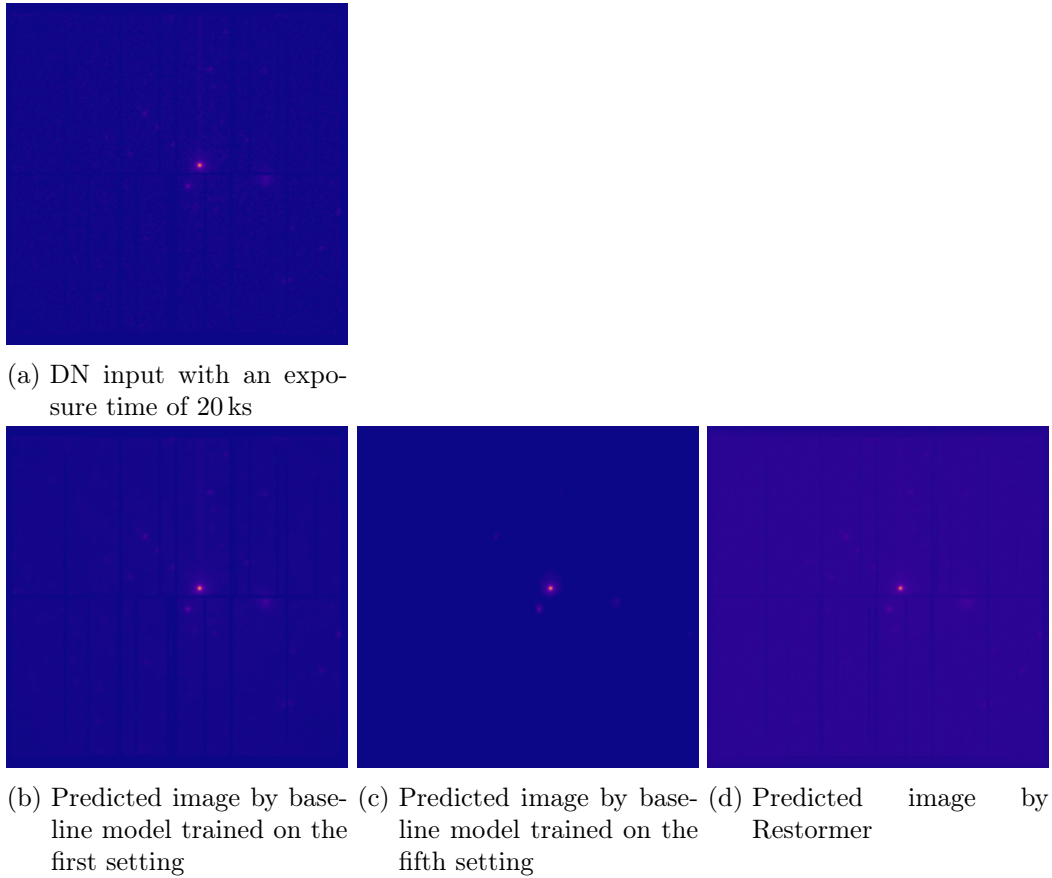


Figure 6.4: DN performance of the models on the real dataset

7 Conclusion and Future Work

In this chapter, we summarize the key achievements of the project and outline potential directions for future work.

To begin, we recreated the dataset used for model training and testing. This new dataset spans the full energy spectrum and includes simulations of individual CCDs. We also simulated the EPIC-MOS detectors, combining them with the EPIC-pn detector using a straightforward, though admittedly naive, approach. Importantly, we avoided manual filtering of dim subhalos, which had not been explicitly defined. Additionally, we updated the detector masks to ensure better accuracy in our simulations. Once the dataset was ready, we retrained the baseline models using the newly generated data. Beyond these baseline models, we also trained more advanced architectures like Restormer and DRCT using different dataset configurations to explore their effectiveness in handling the specific challenges posed by our data.

In conclusion, our simulated dataset enhances the performance of both SR and DN models. Furthermore, employing transformer models like Restormer and DRCT is likely to yield even better and more visually appealing results with extended training.

For future research, several improvements and avenues of exploration have been identified. First, updating the RMF and ARF to reflect the most recent observational data would improve the accuracy of the simulations. Additionally, simulating the EPIC-pn detector with a higher resolution, such as 4x, could yield better results in super-resolution tasks.

In line with recent advancements (e.g., Soltis et al. [122]), we propose assuming a perfect instrument response and PSF. Although the exact implementation details from Soltis' work are unavailable, replicating this approach could significantly improve image quality.

We also identified potential improvements in how we conduct simulations. Rather than simulating detectors to match specific image sizes, a more precise method would be to simulate detectors based on achieving a specific pixel size. For example, to achieve an EPIC-pn 2x (or 4x) resolution, we could simulate EPIC-MOS with a 0.5x (or 1x) scale. Similarly, simulating eROSITA with a 1x scale would provide better alignment with the desired resolutions. Moreover, the current approach for combining multiple detectors is somewhat simplistic, and a more sophisticated method could improve accuracy.

Exploring alternative loss functions could further refine model performance. Given that the dataset consists of relatively small values and the data distribution is Poisson-like, using

a Poisson loss function might be more appropriate for the task, as suggested by Sweere et al. [1].

Additionally, future work could involve training the transformer models, such as Restormer, for a longer duration and using the combined HR images, rather than just the EPIC-pn data. This would provide a more comprehensive training set and potentially better generalization.

Finally, it is crucial to recreate the real dataset used for testing, especially for the SR task, as we currently have only 109 images, assuming an LR exposure of 20 ks and a HR exposure of 100 ks. When recreating the dataset, it will be important to ensure that separate EPIC-MOS images are obtained and combined in the same manner as the simulated images, while also making sure to cover the entire energy spectrum. This would result in a more robust dataset for future testing and validation.

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Acronyms

AI	Artificial Intelligence	HEW	Half Energy Width
AGN	Active Galactic Nucleus	HR	High-Resolution
ALE	Arbitrary Lagrangian-Eulerian	IQA	Image Quality Assessment
AMR	Adaptive Mesh Refinement	ΛCDM	Λ Cold Dark Matter
ARF	Ancillary Response File	LR	Low-Resolution
BKG	Background	MAE	Mean Absolute Error
CCD	Charge-Coupled Device	MCP	Microchannel Plate
CCF	Current Calibration File	MDTA	Multi-Dconv Head Transposed Attention
CNN	Convolutional Neural Network	MISR	Multiple Image Super-Resolution
CMB	Cosmic Microwave Background	MLP	Multilayer Perceptron
CZT	Cadmium Zinc Telluride	MOS	Metal-Oxide-Semiconductor
DL	Deep Learning	MS	Multi-Scaling
DN	Denoising	MSE	Mean Squared Error
DRCT	Dense-Residual-Connected Transformer	MS-SSIM	Multi-Scaling-SSIM
EPIC	European Photon Imaging Camera	NLP	Natural Language Processing
FITS	Flexible Image Transport System	ODF	Observation Data File
FLRW	Friedmann-Lemaître-Robertson-Walker	OM	Optical Monitor
FOV	Field of View	PC	Phase Congruency
FSIM	Feature Similarity Index	PPS	Processing Pipeline Subsystem
FWHM	Full Width Half Maximum	PSF	Point Spread Function
GAN	Generative Adversarial Network	PSNR	Peak Signal-to-Noise Ratio
GDFN	Gated-Dconv Feedforward Network	QE	Quantum Efficiency
HaarPSI	Haar wavelet-based Perceptual Similarity Index	ReLU	Rectified Linear Unit
		RDB	Residual-Dense Block
		RDG	Residual Deep feature extraction Group
		RGS	Reflection Grating Spectrometer

RMF	Redistribution Matrix File	SR	Super-Resolution
RNN	Recurrent Neural Network	SSIM	Structural Similarity Index Measure
RRDB	Residual-in-Residual Dense Block	SSDA	Stacked Sparse DN Autoencoder
SAS	Science Analysis System	STL	Swin Transformer Layer
SDRCB	Swin-Dense-Residual-Connected Block	SWCX	Solar Wind Charge Exchange
SGD	Stochastic Gradient Descent	TNG	The Next Generation
SIMPUT	Simulation Input	UHB	XMM-Newton User Handbook
SISR	Single Image Super-Resolution	XML	Extensible Markup Language
SIXTE	Simulation of X-ray Telescopes	XMM	X-ray Multi Mirror
SPH	Smoothed Particle Hydrodynamics	XSA	XMM-Newton Science Archive

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